

User Manual

Predictive Maintenance Solution

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PREFACE

The user manual provides a detailed description of application functionality (Predictive Maintenance System), which help any user to understand and leverage the solution in best possible way.

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1 INTRODUCTION

The proposed solution is a comprehensive system for managing industrial processes, focusing on authentication, authorization, digitizing check-sheets, centralized data visualization, history tracking, reporting, email notifications, mid-year changes to check-sheets, and data backup. It encompasses user management, role-based access control, check-sheet creation and approval, data visualization, machine history maintenance, report generation, email configuration, check-sheet modifications, and data backup for security and continuity.

2 SYSTEM ARCHITECTURE

There are eight types of users in the system:

1. Admin
2. MTD HOD
3. PRD HOD
4. MTD HOS
5. PRD HOS
6. MTD TL/HOSS
7. PRD TL/HOSS
8. Operator

Following table gives the user permissions to access the various pages:

3 SOFTWARE DESCRIPTION

3.1 Overview

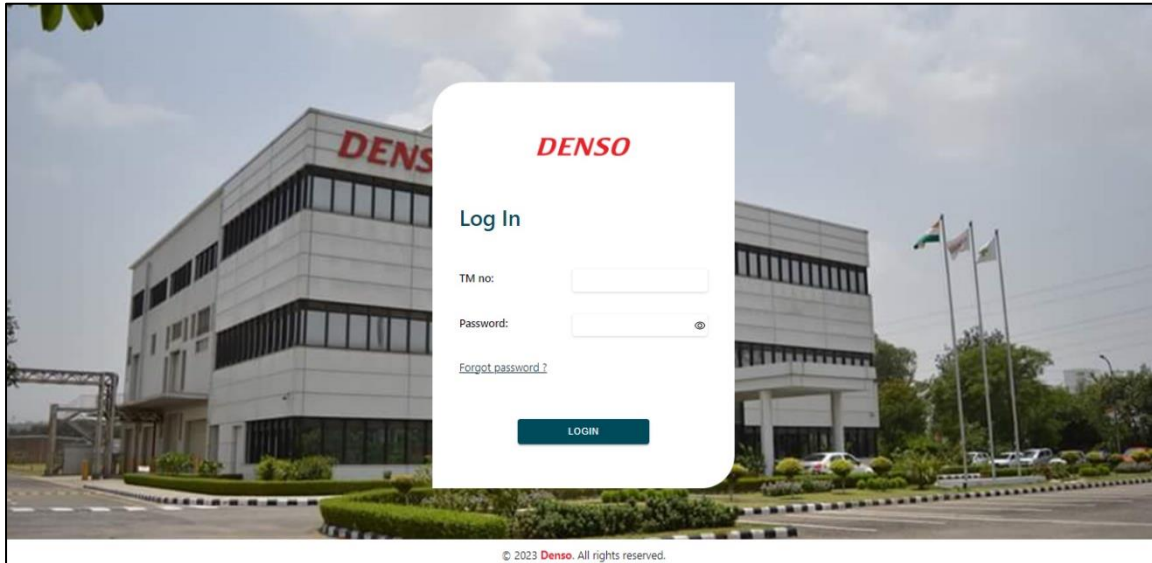
The web application has the following screens/pages:

1. Login
2. Main dashboards
 - a) Plant dashboard
 - b) Section dashboard
3. Creation dashboard
 - a) Check-sheet dashboard
 - b) User-assign
 - c) Check-sheet dashboard
4. Approval
 - a) Preparation approval
 - b) Planning approval

- c) Implementation approval
 - d) Monthly reports(machine wise) approval
 - e) Annual PM schedule approval
- 5. Approval log
 - a) Preparation/ Planning log
 - b) PM Plan vs Actual log
- 6. PM log
 - a) PM log
 - b) Pending PM log history
- 7. PM report
 - a) Monthly report (Machine wise)
 - b) Monthly report(Line wiser)
 - c) Annual PM schedule
 - d) Annual PM vs Actual
 - e) PM Time monitoring
- 8. Back-end data
- 9. Open abnormality tracking
- 10. Spare entry
- 11. Spare log
- 12. Spare report
- 13. Profile

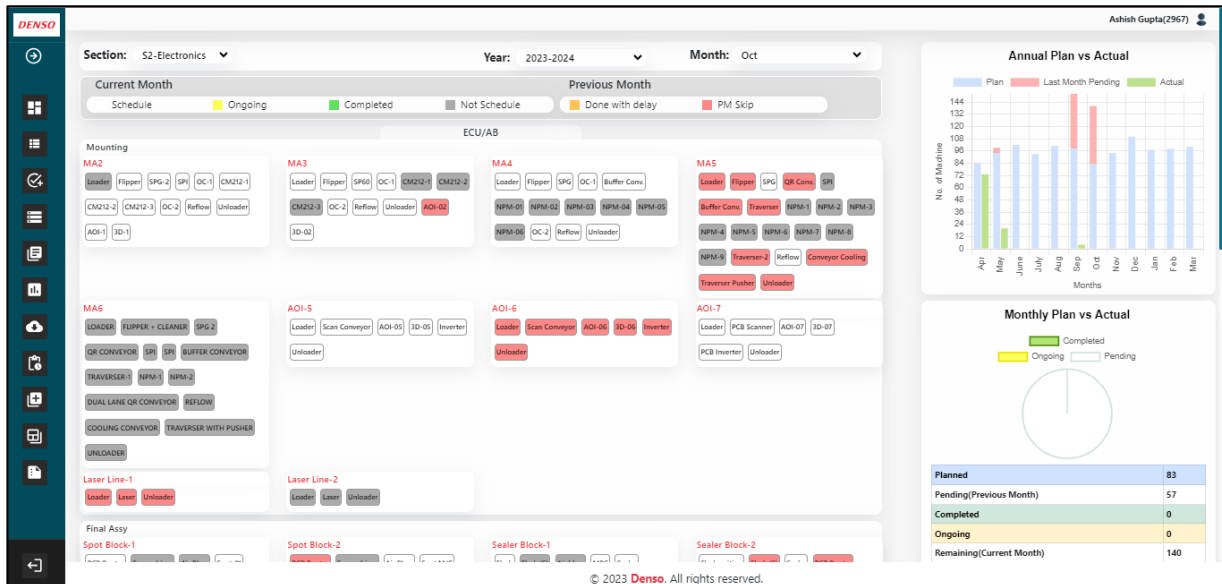
3.2 Web Application

3.2.1 Main Login



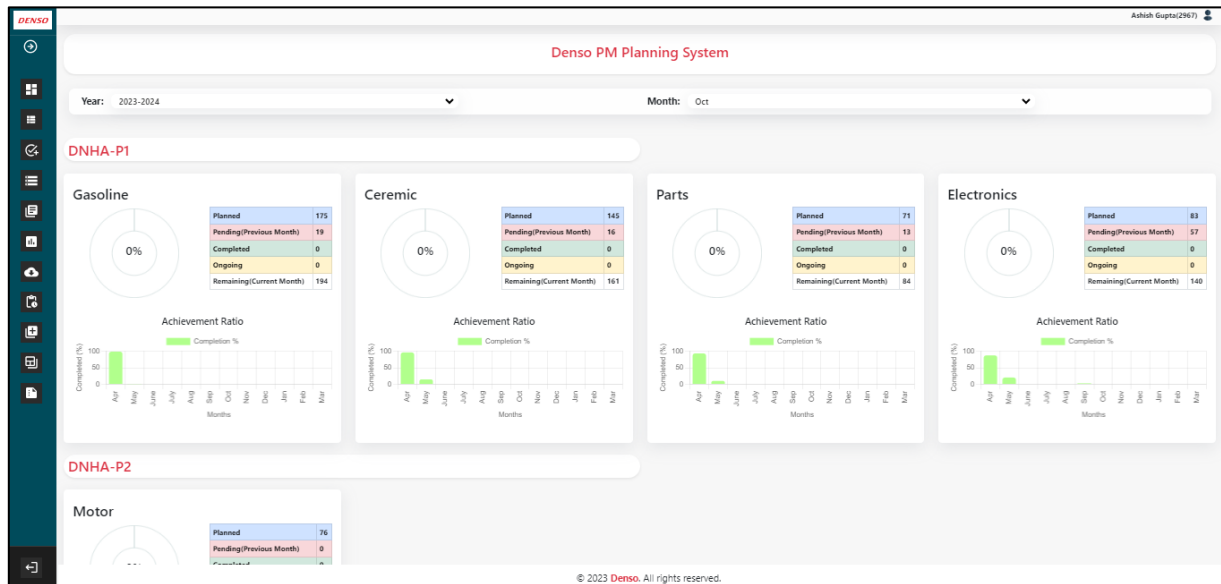
- The GUI shown above is the main login screen used to validate a user for access of dashboard. The user needs to enter the TM no.(Team Member No) and password and then click on “Login” button. If the credentials are correct, the user will be redirected to main dashboard.
- If a user forgets their password, they should enter their registered email address to receive a link to change their password.

3.2.2 Section Dashboard



- This dashboard is common for all user but not every user see check-sheet of other section.
- On selection of machine, check-sheet is open and display every data of the selected machine. On the machine check-sheet data, the access privilege is different based on the users type.
- Every uses take decision based on the machine status color code and also analyze the chart data and made decision.
- As users apply the filter, users can also examine past year and previous month data.
- MTD TL/HOSS add remarks on the displayed machine color code and based on this they take some decision. Other user see this remarks given by MTD TL/HOSS.

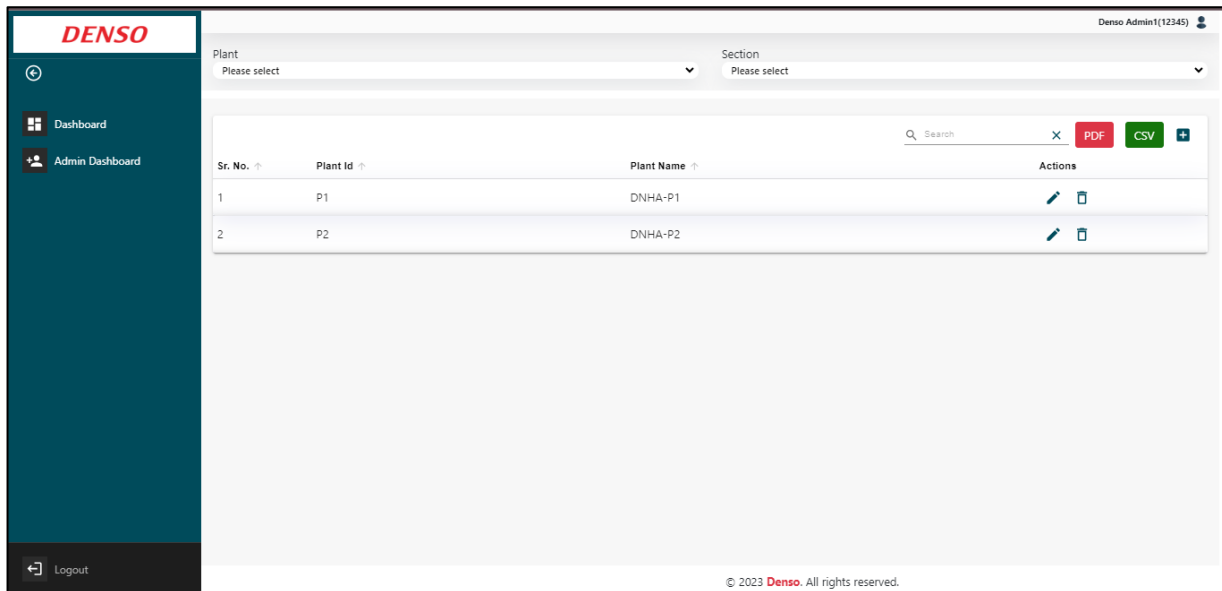
3.2.3 Plant Dashboard



- Plant dashboard contain every registered plants of the company. And display each section data of the plant.
- On this dashboard user take decision regarding the plan vs actual and also display completion in percentage(%) with respect to every month.
- There is one table which show planned, pending (previous month), completed, ongoing, remaining (Current month) with their counts.
- As users apply the filter, users can also examine past year and previous month data.

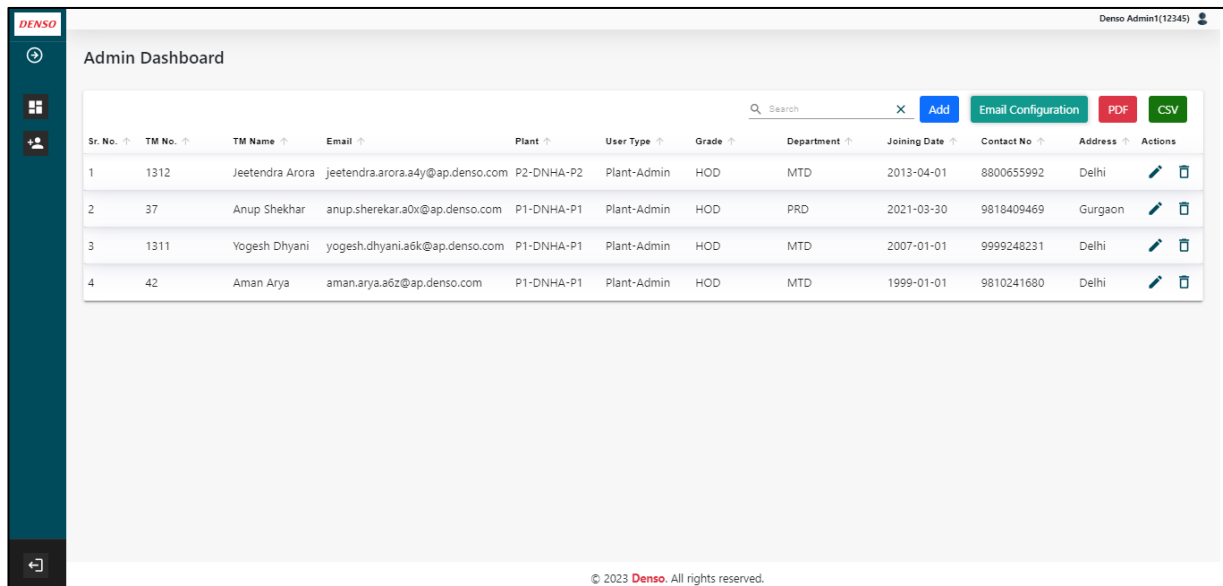
3.2.4 Admin User

3.2.4.1 Plant / Section/ Sub-section creation dashboard

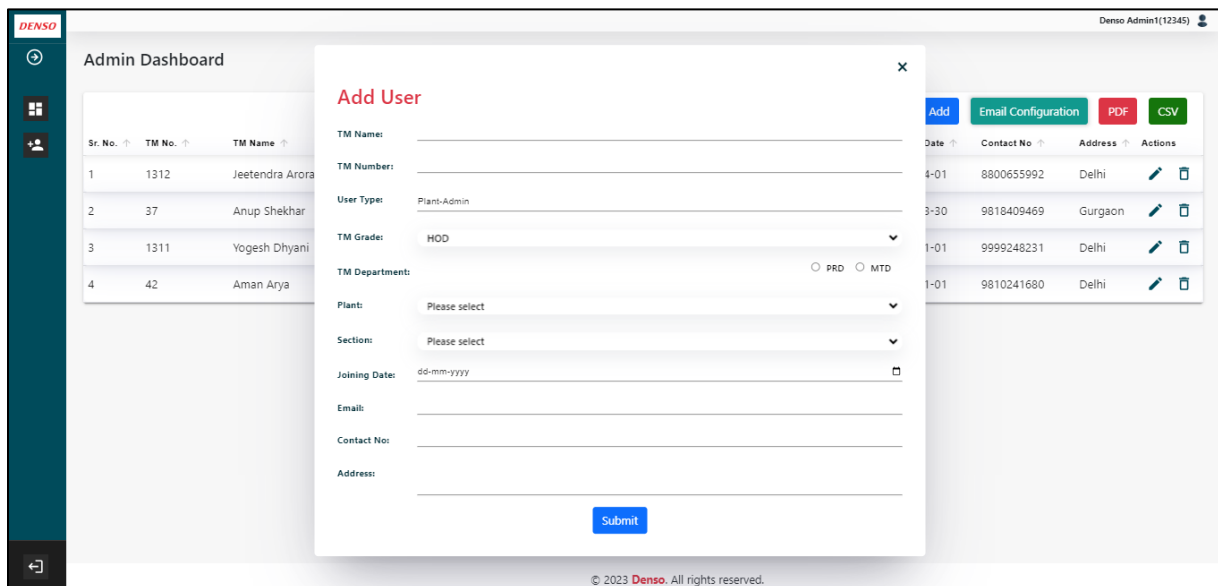


- Admin can add new plant of the company as their requirement. On this dashboard admin will be update and delete company plant.
- Also based on plant selection admin can add new section and it's sub-section.
- When admin add new section for particular plant, they need to select dashboard level (Yer/No). Based on this other user will see their dashboard on this selection.
- Example:
 - On Electronics : If dashboard level is Yes. So use will display dashboard on section level not it's sub-section level.
 - On Powertrain : If dashboard level is No. So use will display dashboard on sub-section(Gasoline, Ceramic, Parts) level not it's section level.
- Admin add sub-section based on the selected section and also update and delete sub-sections. Also, admin will add new sub-section with their sequence.

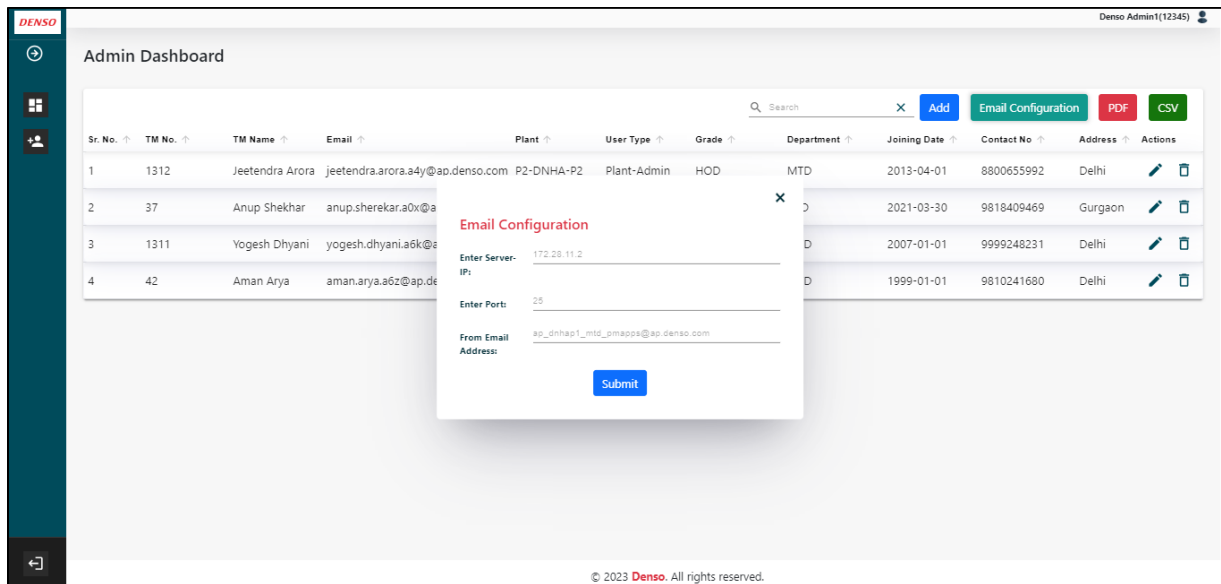
3.2.4.2 Admin Dashboard (User Management)



- Admin will add new user which is Plant-Admin user type and also with grade(HOD/HOS) and department(MTD/ PRD).



- Admin will be able to add new user with assign plan, section, sub-sections and other required details.
- Also user will be able to update and delete Plant-Admin user.



- In email configuration Admin will able change email id, IP address and port which is use in every email send by PM software.

3.2.5 MTD/PRD HOD

3.2.5.1 Creation Dashboard (Sub-section)

The screenshot shows the DENSO Creation Dashboard (Sub-section) interface. The sidebar on the left contains navigation options: Dashboard, Creation, Approval, PM Log, PM Report, Open Abnormality Trac..., Spare Log, and Spare Report. The main area displays a table with columns: Sr No, SubSection Id, SubSection Name, Sequence, and Actions. The table contains three rows of data:

Sr No	SubSection Id	SubSection Name	Sequence	Actions
1	SS4	ECU/AB	1	[Edit] [Delete]
2	SS5	Meter	2	[Edit] [Delete]
3	SS6	RTS	3	[Edit] [Delete]

The interface also includes a search bar, PDF and CSV export buttons, and a user profile dropdown. The footer shows the copyright notice: © 2023 Denso. All rights reserved.

- HOD user can add new section and sub-section with respect to assign plant.
- MTD/PRD HOD add new sub-sections with it's sequence to the particular assign section. Also user will able to update and delete sub-sections.

3.2.5.2 Six-month Implementation Approval Dashboard

Sr No	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	O2	O2 2W Line-01	M-EN-O2-2BA-01-010-1	Inner cover 8 point crimping	Implementation View
2	O2	O2 2W Line-01	M-EN-O2-2WFA-01-170-1	Plus holder	Implementation View
3	O2	O2 2W Line-01	M-EN-O2-2WFA-01-190-1	Plus holder & heater lead wire crimping & welding	Implementation View
4	O2	O2 2W Line-01	M-EN-O2-2WFA-01-200-1	Minus holder & heater lead wire crimping & welding	Implementation View
5	O2	O2 2W Line-01	M-EN-O2-2WFA-01-180-1	Heater p/f	Implementation View
6	Injector	UC#1	EICI-111	Camera Inspection Machine	Implementation View
7	Injector	UC#1	EICI-112	Camera Inspection Machine	Implementation View
8	O2	O2 2W Line-01	M-EN-O2-2BA-01-030-1	Housing welding	Implementation View

- HOD user can see six month (Sep-Mar) approval data on this dashboard with yearly filtering.
- This user will get email for six-month approval when Sep-Mar will come, So user can easily track the approval process.
- On click of implementation button user can see below approval check-sheet and HOD user will be able to approve and give remarks on received check-sheet.

Kindly approve checkSheet of Mar month. ☐ Yes

Remarks:

PLAN ACCEPTANCE (By PRD TL)		PLAN PREPARED (By MTD TL)	
Mukul		vivek yadav	

Year: 2023-2024

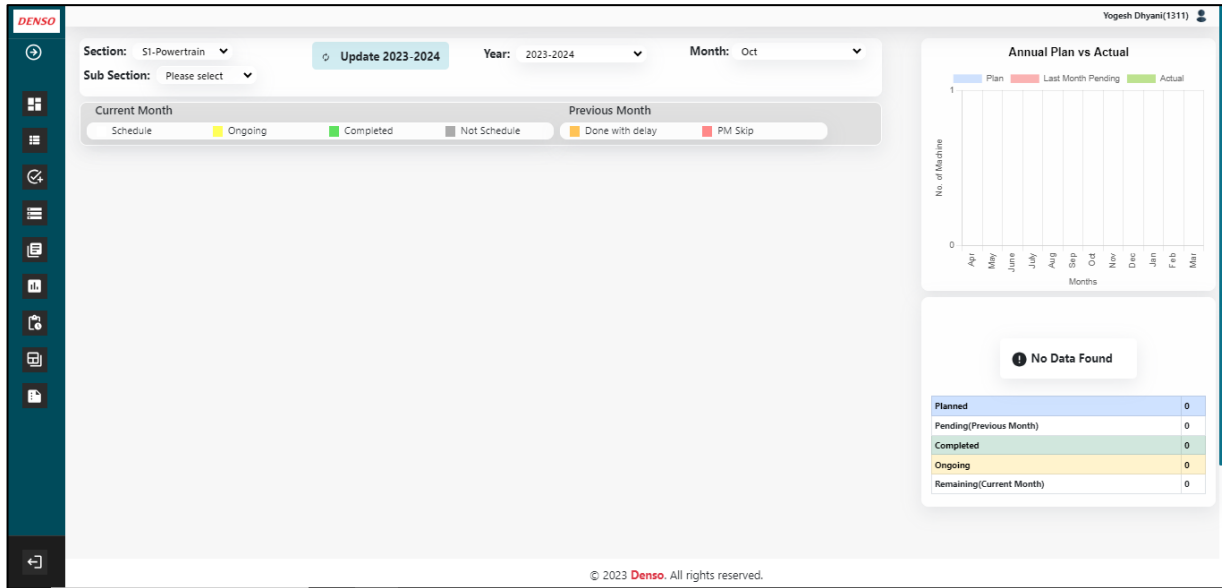
Line- M/c No : M-EN-O2-2BA-01-010-1	Machine Name: Inner cover 8 point crimping	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verified by (MTD TL)	Approved by (MTD HOS)	Approved by (MTD HOD)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	
1	Part Holding Cylinder	Cylinder Movement	It should not hard/damaged	Check / adjust	1/1M	M	10	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
2	Centering	Crimping station centring	By centering jig	Check / adjust	1/1M	M	30	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
3	Poke-yoke check point	As per function sheet	It should give alarm	Check / adjust	1/1M	M	30	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
4	Air Filter	Element	It should not be dirty	Change	1/3M	M	15	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
5	Sol. Manifold	Exhaust silencer	It should not loose / damage	Check / adjust	1/3M	M	30	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
6	Sinto press	Belt Tension & condition	Belt should not be loose & wear out	Check / adjust/Change	1/3M	M	60	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
7	Machine back up	-----	-----	Update machine back up as per back up sheet	1/6M	M	60	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->

3.2.5.3 User Assign Dashboard

The screenshot displays the 'User Assign Dashboard' interface. A modal window titled 'Add User' is open in the center, allowing for the creation of a new user. The modal includes fields for 'TM Name', 'TM Number', 'User Type' (set to 'Section-Admin'), 'TM Grade' (set to 'HOS'), 'Plant' (set to 'P1-DNHA-P1'), 'Section' (a dropdown menu), 'Sub Section' (a dropdown menu), 'Joining Date' (with a date picker), 'Email', 'Contact No.', and 'Address'. A 'Submit' button is located at the bottom of the modal. In the background, the dashboard shows a table of existing users with columns for 'Sr. No.', 'TM No.', and 'TM'. To the right, there is another table with columns for 'Section', 'Sub Section', and 'Actions', listing various vehicle components like 'S2-Electronics' and 'S1-Powertrain'.

- MTD HOD will only be able to add new Section-admin user-type with assigned plant, section, sub-sections.
- Also user will be able to update and delete Section-admin.

3.2.5.4 Financial Year Change



Note:

- When new financial year will come this button is display on the section dashboard on every plants MTD HODs side.
- MTD HOD should be click on this update button, after click on this button every assign plan machine check-sheet will be automatically created and this check-sheet status is on “Planning ” phase.
- If user need to update the planning that will be done on check-sheet dashboard. The remaining flow as same as previous year.
- If user wad deleted some points in check-sheet that will be not display in new financial year.

3.2.6 MTD/PRD HOS

3.2.6.1 Creation Dashboard (Cell/product, Line)

The screenshot shows the DENSO Creation Dashboard interface. The sidebar on the left contains navigation links: Dashboard, Creation, Creation Dashboard, User Assign, Approval, Approval Log, PM Log, PM Report, Back-end Data, Open Abnormality Tr..., Spare Log, and Spare Report. The main content area has a header with 'Sub-section' (SS4-ECU/AB) and 'Cell/Product' (Please select) dropdowns. Below this is a table with the following data:

Sr No	Cell/Product Id	Cell/Product Name	Cell/Product Sequence	Actions
1	C1	Mounting	1	[Edit] [Delete]
2	C2	Final Assy	2	[Edit] [Delete]

At the top right of the table area, there are buttons for PDF, CSV, and a search bar. The footer of the interface shows '© 2023 Denso. All rights reserved.'

- HOS user can add new cell/product and line with respect to assign section and sub-section.
- MTD HOS add new cell/product with it's sequence to the particular assign section and sub-section. Also user will able to update and delete cell/product.
- Based on cell/product selection user will add new line with it's sequence. Also user will able to update and delete line.

3.2.6.2 User Assign Dashboard

The screenshot displays the 'User Assign Dashboard' interface. On the left, a sidebar contains navigation icons. The main area features a table with columns 'Sr. No.', 'TM No.', and 'TM Name'. The table lists 9 users. Overlaid on this is a modal window titled 'Add User'. This modal contains several form fields: 'TM Name', 'TM Number', 'TM Department' (with radio buttons for PRD and MTD), 'Plant' (with a dropdown menu), 'Section' (with a dropdown menu), 'Sub Section' (with a dropdown menu), 'Cell/Product' (with a dropdown menu), 'Joining Date' (with a date picker), 'Email', 'Contact No.', and 'Address'. A 'Submit' button is at the bottom of the modal. In the background, another modal is partially visible, showing a list of products and actions.

Sr. No.	TM No.	TM Name
1	3485	Tushar
2	1178	Sunil Poonia
3	2967	Ashish Gupta
4	89	Chanchal
5	843	Rajeev Gupta
6	954	Gaurav Sharma
7	3419	Vishnu
8	3431	Sandeep Dhan
9	2762	Anurag Bhatt

- MTD HOS will only able to add new user TL/HOSS, Operator user-type with PRD and MTD grade assigned plant, section, sub-sections.
- Also MTD HOS will able to add Section-Admin with only PRD grade.
- Also user will able to update and delete Section-Admin, TL/HOSS, Operator.

3.2.6.3 Preparation Approval dashboard

DENSO

Preparation Approval Dashboard

Ritu Bhatia(899)

Sr No	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-2	EEAT-171	THD DC1U(4830)	<button>Preparation</button> <button>View</button>

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- On this dashboard every preparation approval will display and on click of Preparation button below screen will be open and based this MTD HOS will review and approve the request.

DENSO

Checksheet Data

Ritu Bhatia(899)

Sr. No.	C	Inspection Item	Inspection Point	Judgement Criteria	Action	Cycle	Person In Charge	PM Time (min)	Actions
1	B	Cutting Portion	Blade Movement	Should be smooth without abnormal sound.	Only Clean	1/3M	M	10	
2	B	Cutting Portion	Lead Cutting Length	"Should be with in Spec (2.5-4.5)"	Adjust	1/3M	M	30	
3	B	Cutting Portion	Blade Condition	No Damage	Inform PSG for Replacement	1/3M	M	30	
4	B	Positioning Pin	Positioning pin Movement	Movement Should Be Smooth	Only Clean	1/3M	M	15	
5	B	Positioning Pin	Positioning pin Loosenes	Positioning Pin Bolt Not Loose	Tightning	1/3M	M	30	
6	B	Rail	Condition Check	NO Wear And Tear	Repair or Replace.	1/3M	M	20	
7	B	Limit Switch	ON/OFF	Should Work Properly	Check/Change	1/3M	M	10	
8	B	Motor	Motor	Running smoothly.No abnormal sound	Repair or Replace.	1/3M	M	30	
9	B	Lubrication	All Movement Parts	Lubrication Apply	Clean And Lubrication	1/3M	M	20	

View & Send for Approval

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- After reviewing every data of preparation phase MTD HOS will approve/reject the request as below display.

- If MTD HOS need to approve the request then select Yes and submit, else reject then select No and enter relatable remarks (required).

Ritu Shafia(999)

Kindly approve checkSheet. ☐ Yes ☒ No

Remarks:

Submit

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

Line:- M/c No : EEAT-171	Machine Name: THD DC1U(4830)	Approved by (MTD HOD)	Prepared by (MTD TL)	Checked & Verify by (MTD TL)															
		Ashish Gupta	Sreeranth	Approved by (MTD HOD)															
				Approved by (MTD HOD)															
SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	PM Time Charge	Initial	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan
1	B	Cutting Portion	Blade Movement	Should be smooth without abnormal sound.	Only Clean	1/3M	M	10											
2	B	Cutting Portion	Lead Cutting Length	"Should be with In Spec (2.5-4.5)"	Adjust	1/3M	M	30											
3	B	Cutting Portion	Blade Condition	No Damage	Inform PSS for Replacement	1/3M	M	30											
4	B	Positioning Pin	Positioning pin Movement	Movement Should Be Smooth	Only Clean	1/3M	M	15											
5	B		Positioning pin Loose	Positioning Pin Bolt Not Loose	Tightning	1/3M	M	30											
6	B	Rail	Condition Check	NO Wear And Tear	Repair or Replace.	1/3M	M	20											
7	B	Limit Switch	ON/OFF	Should Work Properly	Check/Change	1/3M	M	10											

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3.2.6.4 Implementation Approval Dashboard

Sr No	Month	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	Apr	Mounting	MA2	EETP-094	SPG-2	Implementation View
2	Apr	Mounting	MA2	EETP-013	OC-1	Implementation View
3	Apr	Mounting	MA2	EETP-044	SPI	Implementation View
4	Apr	Mounting	MA2	EETP-017	OC-2	Implementation View
5	Apr	Mounting	MA3	EETP-033	YS Loader	Implementation View
6	Apr	Mounting	MA3	EETP-036	OC-1	Implementation View
7	Apr	Mounting	MA3	EETP-040	OC-2	Implementation View
8	Apr	Mounting	MA4	EETP-052	Board Flipper	Implementation View
9	Apr	Mounting	MA4	EETP-057	Multi-Machine Loader	Implementation View

- On Implementation approval dashboard receive approval request of implementation work done by operator.
- This request only show when MTD TL was approved.
- On click of implementation button user will display below approval screen.

Sr No	Month	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	Apr	Mounting	MA2	EETP-094	SPG-2	Implementation View
2	Apr	Mounting	MA2	EETP-013	OC-1	Implementation View
3	Apr	Mounting	MA2	EETP-044	SPI	Implementation View
4	Apr	Mounting	MA2	EETP-017	OC-2	Implementation View
5	Apr	Mounting	MA3	EETP-033	YS Loader	Implementation View
6	Apr	Mounting	MA3	EETP-036	OC-1	Implementation View
7	Apr	Mounting	MA3	EETP-040	OC-2	Implementation View
8	Apr	Mounting	MA4	EETP-052	Board Flipper	Implementation View
9	Apr	Mounting	MA4	EETP-057	Multi-Machine Loader	Implementation View

Kindly approve checkSheet. ☐ Yes ☐ No

Submit

PLAN ACCEPTANCE (By PRD TL)	PLAN PREPARED (By MTD TL)
Harsh	Ashish Gupta

Year: 2023-2024

Ashish Gupta																			

Line-
M/c No : EETP-094

Machine Name: SPG-2

Approved by
(MTD HOS)

Ritu Bhatia

Prepared by
(MTD TL)

Anshul Gupta

Checked & Verify by
(MTD TL)

Approved by
(MTD HOS)

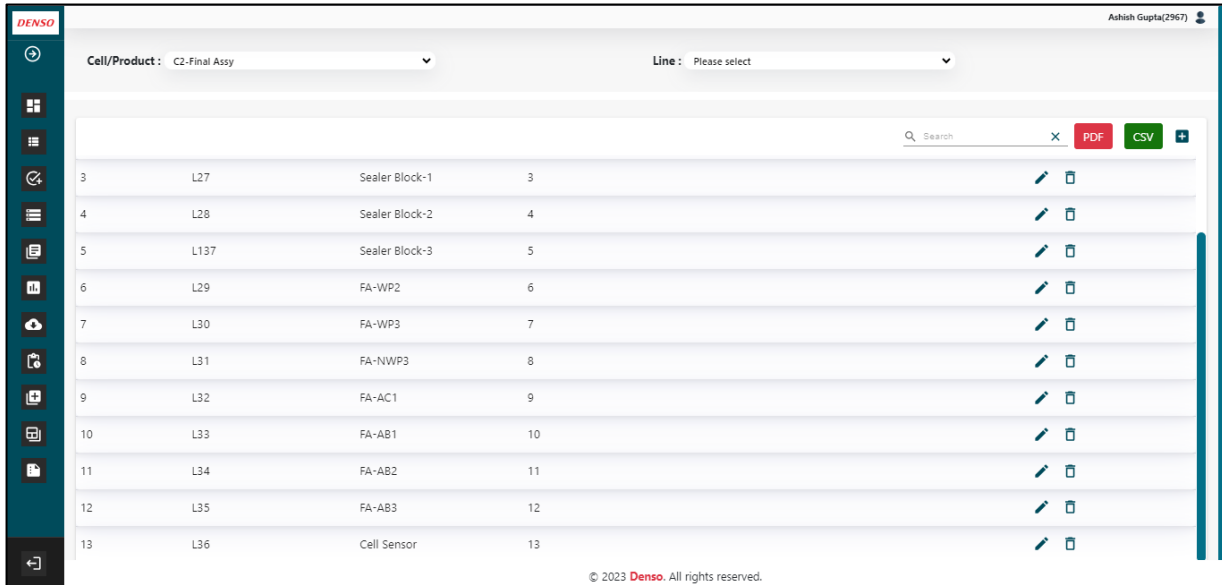
Approved by
(MTD HOD)

SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	PM Time change (min)	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan
1	B	X-Y Table & Inlet/outlet	Conveyor belt	Should not wear/ tear	Clean/ replace	1/1M	M	60	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
2	B	PCB Recognition camera	Camera	Should not dirty/proper functioning	Clean/replace	1/1M	M	20	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
3	B	Solder Detection Sensor	Sensor	Should not dirty/proper functioning/Check input Screen of machine	Clean/replace	1/1M	M	20	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
4	B	PCB Detection Sensor	Sensor	Should not dirty/proper functioning/Check input Screen of machine	Clean/replace	1/1M	M	30	-->	-->	-->	-->	-->	-->	-->	-->	-->	-->
5	B	All Sensor/Safety Switch	Sensor and safety switch	Should not dirty/proper functioning/Check input Screen of machine	Clean/replace	1/3M	M	30		-->			-->	-->		-->		
6	B	Air regulator	Filter	Should not dirty/visual check	Replace	1/3M	M	30		-->			-->	-->		-->		
7	B	CPU Box	Filter	Should not dirty/visual	Replace/Clean	1/3M	M	30		-->			-->	-->		-->		

- The Implementation approval process is same as preparation approval. Just need to review every filled data by operator and color indication (Red - skip or no completion, Orange - done with delay).
- If MTD HOS reject this request, operator need to work on rejected remarks and again send approval as flow of approval.

3.2.7 MTD TL/Hoss

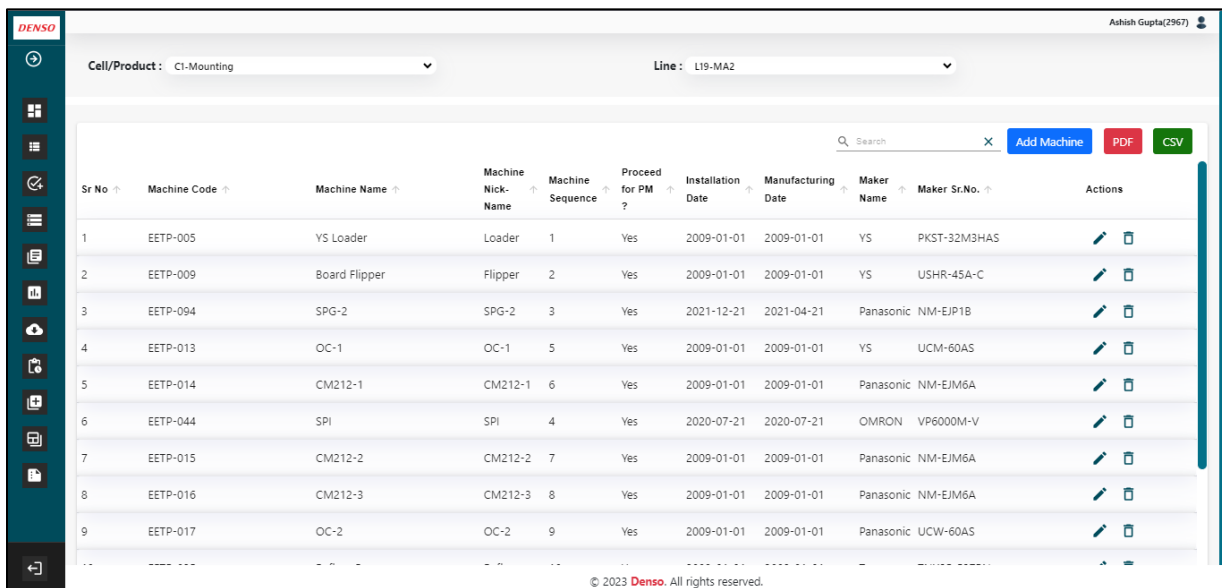
3.2.7.1 Creation Dashboard (Lines/Machines)



The screenshot shows the 'Creation Dashboard (Lines/Machines)' interface. At the top, there are dropdown menus for 'Cell/Product' (set to 'C2-Final Assy') and 'Line' (set to 'Please select'). Below these is a search bar and buttons for 'PDF' and 'CSV'. The main table lists 13 lines, each with a sequence number, a line number, a machine name, and a machine number. Each row has edit and delete icons.

Line No.	Line	Machine Name	Machine No.
3	L27	Sealer Block-1	3
4	L28	Sealer Block-2	4
5	L137	Sealer Block-3	5
6	L29	FA-WP2	6
7	L30	FA-WP3	7
8	L31	FA-NWP3	8
9	L32	FA-AC1	9
10	L33	FA-AB1	10
11	L34	FA-AB2	11
12	L35	FA-AB3	12
13	L36	Cell Sensor	13

- MTD TL/Hoss user can add new line and machine with respect to assign section, sub-section, cell/product.
- Based on cell/product selection user will add new line with it's sequence. Also user will be able to update and delete line.



The screenshot shows the 'Creation Dashboard (Machines)' interface. At the top, there are dropdown menus for 'Cell/Product' (set to 'C1-Mounting') and 'Line' (set to 'L19-MA2'). Below these is a search bar and buttons for 'Add Machine', 'PDF', and 'CSV'. The main table lists 9 machines, each with a serial number, machine code, machine name, machine nick-name, machine sequence, proceed for PM, installation date, manufacturing date, maker name, maker serial number, and actions (edit/delete).

Sr No	Machine Code	Machine Name	Machine Nick-Name	Machine Sequence	Proceed for PM ?	Installation Date	Manufacturing Date	Maker Name	Maker Sr.No.	Actions
1	EETP-005	YS Loader	Loader	1	Yes	2009-01-01	2009-01-01	YS	PKST-32M3HAS	
2	EETP-009	Board Flipper	Flipper	2	Yes	2009-01-01	2009-01-01	YS	USHR-45A-C	
3	EETP-094	SPG-2	SPG-2	3	Yes	2021-12-21	2021-04-21	Panasonic	NM-EJP1B	
4	EETP-013	OC-1	OC-1	5	Yes	2009-01-01	2009-01-01	YS	UCM-60AS	
5	EETP-014	CM212-1	CM212-1	6	Yes	2009-01-01	2009-01-01	Panasonic	NM-EJM6A	
6	EETP-044	SPI	SPI	4	Yes	2020-07-21	2020-07-21	OMRON	VP6000M-V	
7	EETP-015	CM212-2	CM212-2	7	Yes	2009-01-01	2009-01-01	Panasonic	NM-EJM6A	
8	EETP-016	CM212-3	CM212-3	8	Yes	2009-01-01	2009-01-01	Panasonic	NM-EJM6A	
9	EETP-017	OC-2	OC-2	9	Yes	2009-01-01	2009-01-01	Panasonic	UCW-60AS	

- The above screen display machine based on the cell and line selection.

The screenshot displays the 'Add Machine' form in the Predictive Maintenance System. The form is titled 'Add Machine' and includes fields for Machine Code, Machine Name, Machine Nick-Name, Machine Sequence, Proceed for PM? (Yes/No), Installation Date, Manufacturing Date, Maker Name, and Maker Sr.No. There are 'Submit' and 'Reset' buttons at the bottom of the form. To the left of the form is a table with columns 'Sr No' and 'Machine Code', listing machines EETP-005 through EETP-017. To the right of the form is a table with columns 'Maker Sr.No.' and 'Actions', listing machines KST-32M3HAS, SHR-45A-C, IM-EJP1B, CM-60AS, IM-EJM6A, P6000M-V, IM-EJM6A, IM-EJM6A, and CW-60AS. The interface also shows a sidebar with various icons and a top header with the Denso logo and user information.

- User will be add new machine on click of Add machine button below popup display.
- User need to enter every data of machine.
- The selection of proceed for PM(Yes/No), if user select Yes then user will display machine on dashboard and need to work on PM, else No then user not display any where on the dashboard.
- On click of reset, every fill data will be clear from the add machine form.
- Also, user will able to update and delete machine.

3.2.7.2 User Assign Dashboard

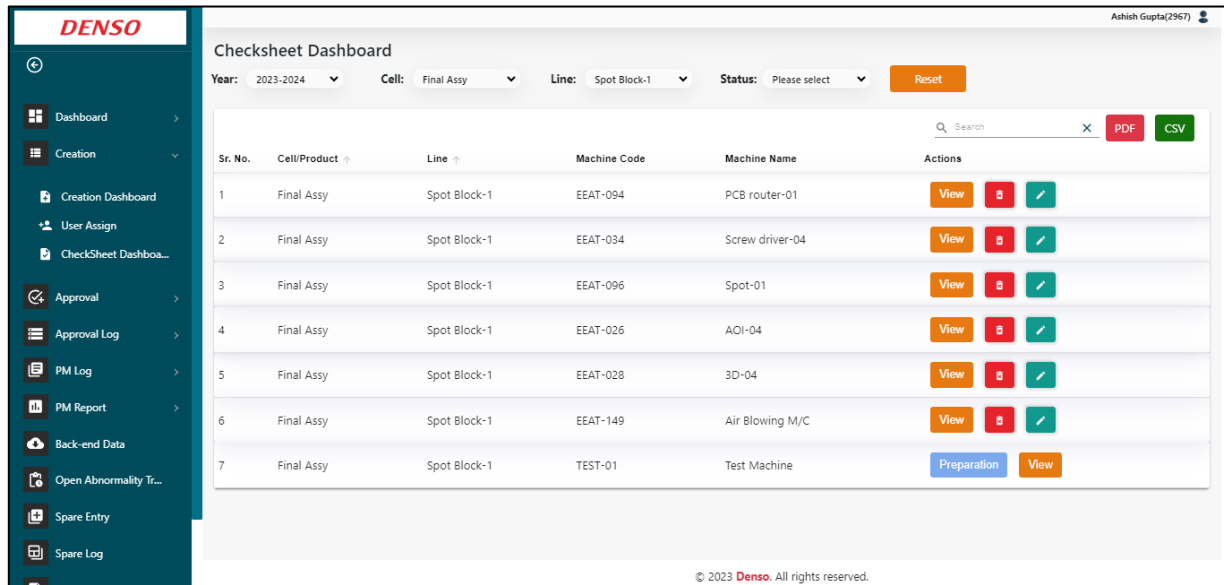
The screenshot displays the 'User Assign Dashboard' interface. On the left, a sidebar contains navigation icons. The main area features a table with 10 rows of user data. A modal window titled 'Add User' is open in the center, containing a form with fields for TM Name, TM Number, User Type (with a dropdown menu showing 'Please select', 'TL/HOSS', and 'Operator'), Plant, Section, Sub Section, Cell/Product, Joining Date, Contact No., and Address. A 'Submit' button is at the bottom of the modal. In the background, a table with columns for 'Sr. No.', 'TM No.', and 'TM Name' is visible, along with an 'Actions' column containing edit and delete icons. The top right corner shows the user 'Ashish Gupta(2967)'.

Sr. No.	TM No.	TM Name
1	3485	Tushar
2	89	Chanchal
3	954	Gaurav Sharma
4	3419	Vishnu
5	2762	Anurag Bhatt
6	566	Mangat
7	3235	Abhishek Swa
8	2973	Harsh
9	865	Shamsheel
10	595	Dheeraj

- MTD TL/Hoss will only able to add new user PRD TL/HOSS, Operator of assigned plant, section.
- User will assign sub-sections and cell/product to the particular user.
- Also user will able to update and delete PRD TL/HOSS, Operator.

3.2.8 Check-sheet Dashboard

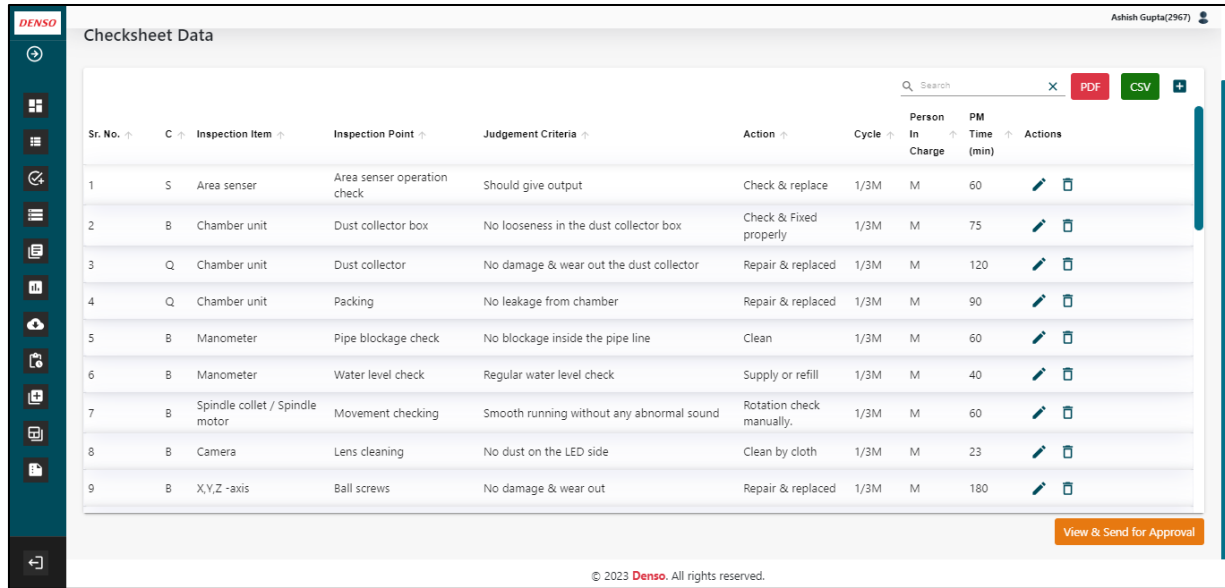
3.2.8.1 Preparation Phase



The screenshot displays the Denso Checksheet Dashboard. On the left is a dark blue sidebar with the Denso logo and a navigation menu including Dashboard, Creation, Approval, PM Log, PM Report, Back-end Data, Open Abnormality Tr..., Spare Entry, and Spare Log. The main content area is titled 'Checksheet Dashboard' and features filters for Year (2023-2024), Cell (Final Assy), Line (Spot Block-1), and Status (Please select), along with a Reset button. Below the filters is a table with columns: Sr. No., Cell/Product, Line, Machine Code, Machine Name, and Actions. The table lists seven machines, all in the 'Preparation' status. The last row for 'Test Machine' has 'Preparation' and 'View' buttons. At the top right of the table area are search, PDF, and CSV export buttons. The footer shows '© 2023 Denso. All rights reserved.'

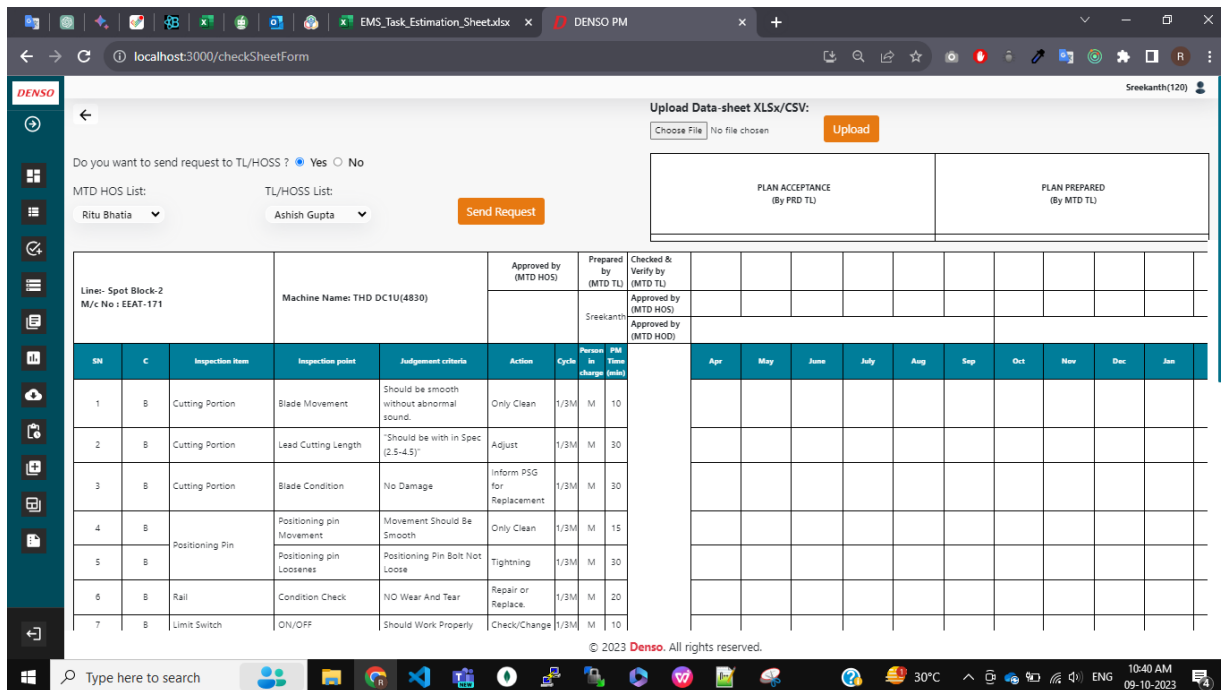
Sr. No.	Cell/Product	Line	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-1	EEAT-094	PCB router-01	View [icon] [icon]
2	Final Assy	Spot Block-1	EEAT-034	Screw driver-04	View [icon] [icon]
3	Final Assy	Spot Block-1	EEAT-096	Spot-01	View [icon] [icon]
4	Final Assy	Spot Block-1	EEAT-026	AOI-04	View [icon] [icon]
5	Final Assy	Spot Block-1	EEAT-028	3D-04	View [icon] [icon]
6	Final Assy	Spot Block-1	EEAT-149	Air Blowing M/C	View [icon] [icon]
7	Final Assy	Spot Block-1	TEST-01	Test Machine	Preparation View

- After adding new machine and selected proceed for PM - Yes then the machine will display on this as preparation phase.
- In this dashboard machine will display in every phase(Preparation, Planning and Implementation) of cycle.
- Also user can filter data based on cell/product, line and status selection. So that user easily identify and find machines and users can also examine past year data based on year selection.
- Firstly machine check-sheet status will on "Preparation". On click of preparation button the below screen will be open and user will able to add new data or directly copy data from other lines machine which is already created or deleted.
- If user want to add new data directly from the already created or deleted check-sheet then select first select line and then choose already created or deleted, after choose the machine based on your machine criteria and submit.
- The data will automatically copy from selected machine and also user can able to add new data after copied data.



Sr. No.	C	Inspection Item	Inspection Point	Judgement Criteria	Action	Cycle	Person In Charge	PM Time (min)	Actions
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60	
2	B	Chamber unit	Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75	
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120	
4	Q	Chamber unit	Packing	No leakage from chamber	Repair & replaced	1/3M	M	90	
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60	
6	B	Manometer	Water level check	Regular water level check	Supply or refill	1/3M	M	40	
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually.	1/3M	M	60	
8	B	Camera	Lens cleaning	No dust on the LED side	Clean by cloth	1/3M	M	23	
9	B	X,Y,Z -axis	Ball screws	No damage & wear out	Repair & replaced	1/3M	M	180	

- If user want to add new data without any other machine check-sheet selection, then user click on the plus button which is for add new data.
- Data is exists in machine check-sheet then it goes on “Under-Preparation stage”.
- After adding new data click on view and send for approval button the below screen will be display and user will able to send approval to the TL/Hoss or MTD HOS.



SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person In Charge	PM Time (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
1	B	Cutting Portion	Blade Movement	Should be smooth without abnormal sound.	Only Clean	1/3M	M	10										
2	B	Cutting Portion	Lead Cutting Length	"Should be with in Spec (2.5-4.5)"	Adjust	1/3M	M	30										
3	B	Cutting Portion	Blade Condition	No Damage	Inform PSG for Replacement	1/3M	M	30										
4	B	Positioning Pin	Positioning pin Movement	Movement Should Be Smooth	Only Clean	1/3M	M	15										
5	B	Positioning Pin	Positioning pin Bolt Not Loose	Positioning Pin Bolt Not Loose	Tightening	1/3M	M	30										
6	B	Rail	Condition Check	NO Wear And Tear	Repair or Replace.	1/3M	M	20										
7	B	Limit Switch	ON/OFF	Should Work Properly	Check/Change	1/3M	M	10										

- If user want send approval to other TL/Hoss and after MTD HOS , then user need to choose Yes and select related section MTD TL/Hoss from the given list, else user choose NO and directly send request to the MTD HOS .

Upload Data-sheet XLSx/CSV:

Choose File No file chosen Upload

Do you want to send request to TL/HOSS ? ☐ Yes ☒ No

MTD HOS List:

Ritu Bhatia

Send Request

SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	PM Time (min)
1	B	Cutting Portion	Blade Movement	Should be smooth without abnormal sound.	Only Clean	1/3M	M	10
2	B	Cutting Portion	Lead Cutting Length	"Should be with in Spec (2.5-4.5)"	Adjust	1/3M	M	30
3	B	Cutting Portion	Blade Condition	No Damage	Inform PSG for Replacement	1/3M	M	30
4	B	Positioning Pin	Positioning pin Movement	Movement Should Be Smooth	Only Clean	1/3M	M	15
5	B	Positioning Pin	Positioning pin Loosens	Positioning Pin Bolt Not Loose	Tightening	1/3M	M	30
6	B	Rail	Condition Check	NO Wear And Tear	Repair or Replace.	1/3M	M	20
7	B	Limit Switch	ON/OFF	Should Work Properly	Check/Change	1/3M	M	10

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- After selection of MTD TL/Hoss, user should be select MTD HOS from the given list and then click on send request.
- On click of send request, if user select MTD TL/Hoss then selected TL/Hoss will receive mail regarding the approval request.
- After sending approval the machine check-sheet status will go on "Preparation Under Approval".

3.2.8.2 Preparation approval dashboard

The screenshot shows the 'Preparation Approval Dashboard' in the Denso system. The dashboard has a sidebar with navigation options: Dashboard, Creation, Approval, Preparation Approval, Implementation App..., Approval Log, PM Log, PM Report, Back-end Data, Open Abnormality Tr..., Spare Entry, Spare Log, and Spare Report. The main area displays a table with the following data:

Sr No	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-2	EEAT-171	THD DC1U(4830)	<button>Preparation</button> <button>View</button>

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- On this dashboard every preparation approval will display and on click of Preparation button below screen will be open and based this MTD TL/Hoss will review and approve the request.

The screenshot shows the 'Checksheet Data' in the Denso system. The checksheet has a sidebar with navigation options: Dashboard, Creation, Approval, Preparation Approval, Implementation App..., Approval Log, PM Log, PM Report, Back-end Data, Open Abnormality Tr..., Spare Entry, Spare Log, and Spare Report. The main area displays a table with the following data:

Sr. No.	C	Inspection Item	Inspection Point	Judgement Criteria	Action	Cycle	Person In Charge	PM Time (min)	Actions
1	B	Cutting Portion	Blade Movement	Should be smooth without abnormal sound.	Only Clean	1/3M	M	10	
2	B	Cutting Portion	Lead Cutting Length	"Should be with in Spec (2.5~4.5)"	Adjust	1/3M	M	30	
3	B	Cutting Portion	Blade Condition	No Damage	Inform PSG for Replacement	1/3M	M	30	
4	B	Positioning Pin	Positioning pin Movement	Movement Should Be Smooth	Only Clean	1/3M	M	15	
5	B	Positioning Pin	Positioning pin Loosenes	Positioning Pin Bolt Not Loose	Tightning	1/3M	M	30	
6	B	Rail	Condition Check	NO Wear And Tear	Repair or Replace.	1/3M	M	20	
7	B	Limit Switch	ON/OFF	Should Work Properly	Check/Change	1/3M	M	10	
8	B	Motor	Motor	Running smoothly, No abnormal sound	Repair or Replace.	1/3M	M	30	
9	B	Lubrication	All Movement Parts	Lubrication Apply	Clean And Lubrication	1/3M	M	20	

View & Send for Approval

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- After reviewing every data of preparation phase MTD TL/Hoss will approve/reject the request as below display.

←

Kindly approve checkSheet. ☒ Yes ☐ No

Submit

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

Line:	M/c No :	Machine Name:	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verified by (MTD TL)													
M/c No : TEST-01		Test Machine		Ashish Gupta														
SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	PM Time change (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60										
2	B	Chamber unit	Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75										
3	Q		Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120										
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90										
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60										
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40										
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60										
8	B	Camera	Lens cleaning	No dust on the LED	Clean by cloth	1/3M	M	23										

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- If MTD TL/Hoss need to approve the request then select Yes and submit, else reject then select No and enter relatable remarks (required).

←

Kindly approve checkSheet. ☐ Yes ☒ No

Remarks:

Submit

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

Line:	M/c No :	Machine Name:	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verified by (MTD TL)													
M/c No : TEST-01		Test Machine		Ashish Gupta														
SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	PM Time change (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60										
2	B	Chamber unit	Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75										
3	Q		Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120										
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90										
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60										
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40										
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60										

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3.2.8.3 Planning Phase

Checksheet Dashboard

Year: 2023-2024 Cell: Final Assy Line: Spot Block-1 Status: Please select Reset

Sr. No.	Cell/Product	Line	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-1	EEAT-094	PCB router-01	View Plan Edit
2	Final Assy	Spot Block-1	EEAT-034	Screw driver-04	View Plan Edit
3	Final Assy	Spot Block-1	EEAT-096	Spot-01	View Plan Edit
4	Final Assy	Spot Block-1	EEAT-026	AOI-04	View Plan Edit
5	Final Assy	Spot Block-1	EEAT-028	3D-04	View Plan Edit
6	Final Assy	Spot Block-1	EEAT-149	Air Blowing M/C	View Plan Edit
7	Final Assy	Spot Block-1	TEST-01	Test Machine	Planning View Edit

- After approval of preparation phase by MTD HOS, the machine check-sheet will go on “Planning” Phase.
- When the user will click on planning button below screen will be display. It will display data of the check-sheet.

Planning Phase

Line Name: Spot Block-1 Machine Name: Test Machine Machine No: TEST-01

Sr. No.	C	Inspection Item	Inspection Point	Judgement Criteria	Action	Cycle	Person In Charge	PM Time	Start Month	Actions
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60	Please select	✓ X
2	B	Chamber unit	Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75		
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120		
4	Q	Chamber unit	Packing	No leakage from chamber	Repair & replaced	1/3M	M	90		
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60		
6	B	Manometer	Water level check	Regular water level check	Supply or refill	1/3M	M	40		
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60		
8	B	Camera	Lens cleaning	No dust on the LED side	Clean by cloth	1/3M	M	23		
9	B	X,Y,Z -axis	Ball screws	No damage & wear out	Repair & replaced	1/3M	M	180		

- On this screen user can plan the every check-sheet points of the machine.
- The user should be select start month of the particular point and based the start month and cycle the actual plan display on the check-sheet with every next frequency of the cycle.

DENSO

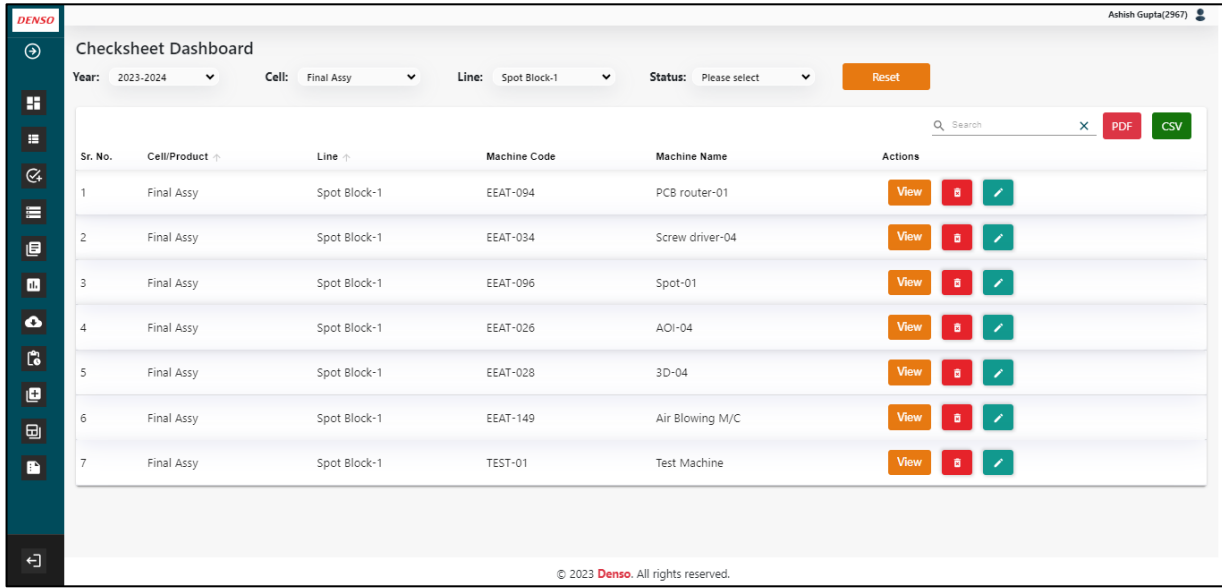
</

- After some selection of the start month the check-sheet status will go on “Under Planning”
- On click of view and send for approval, choose the PRD TL/Hoss for send approval.

DENSO

</

- After sending approval the machine check-sheet status will go on “Planning Under Approval”.



Checksheet Dashboard

Year: 2023-2024 Cell: Final Assy Line: Spot Block-1 Status: Please select Reset

Search PDF CSV

Sr. No.	Cell/Product	Line	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-1	EEAT-094	PCB router-01	View Edit Delete
2	Final Assy	Spot Block-1	EEAT-034	Screw driver-04	View Edit Delete
3	Final Assy	Spot Block-1	EEAT-096	Spot-01	View Edit Delete
4	Final Assy	Spot Block-1	EEAT-026	AOI-04	View Edit Delete
5	Final Assy	Spot Block-1	EEAT-028	3D-04	View Edit Delete
6	Final Assy	Spot Block-1	EEAT-149	Air Blowing M/C	View Edit Delete
7	Final Assy	Spot Block-1	TEST-01	Test Machine	View Edit Delete

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- After approval of planning phase by PRD TL/Hoss, the machine check-sheet will go on “Implementation” Phase.
- When machine check-sheet will go on implementation phase the will display as view action.
- After the implementation phase the Operator user will start the actual work on the machines check-sheet.
- On deletion of machine check-sheet, it will go under the backup data dashboard. So, if user will able to take data from the deletion data to the new check-sheet data.

3.2.9 Open Abnormality Tracking

The screenshot shows the 'Open Abnormality Tracking' dashboard in the Denso system. The sidebar on the left contains navigation links: Dashboard, Creation, Approval, Approval Log, PM Log, PM Report, Back-end Data, Open Abnormality Trac..., Spare Entry, Spare Log, and Spare Report. The main content area has a title 'Open Abnormality Tracking' and filters for 'Year: 2023-2024' and 'Line: Please select'. Below the filters is a table with the following data:

Sr. No.	Line Name	Machine Code	Machine Name	Schedule Month	Checked By	Abnormality Remarks	Target Date	Action Details	Done Date	Done By	Actions
1	MA2	EETP-025	Reflow Oven	Apr	Gagandeep	Filter damaged but spare not available	2023-04-30				<button>Close</button>

- This dashboard show only this data which NG (Not good) while implementation phase.
- **Note:** Only MTD TL/Hoss will able to close this abnormality. First need to set target on click of edit, After completion of work user will need to edit action detail, done date, and done by user then after click on the close button.
- This will be display on check-sheet with dark circle and done date.

3.2.10 Spare Entry

- This dashboard for add spare entry for other module (BM, CM, Kaizen).
- This dashboard entry will be display on spare log.
- **Note:** Spare entry dashboard is only accessible to the MTD TL/Hoss and Operator. Other user will only see spare entry log.

3.2.11 PRD TL/HOSS

3.2.11.1 Planning Approval Dashboard

Sr No	Cell Name	Line Name	Machine Code	Machine Name	Actions
1	Mounting	MA4	EETP-093	NPM-01	View
2	Mounting	MA4	EETP-101	NPM-03	View
3	Mounting	MA4	EETP-056	NPM-04	View
4	Mounting	MA4	EETP-057	NPM-06	View
5	Mounting	MA5	EETP-080	NPM-6	View
6	Mounting	MA5	EETP-096	NPM-7	View
7	Mounting	MA5	EETP-097	NPM-8	View
8	Mounting	MA5	EETP-081	NPM-9	View

- On this dashboard every planning approval will display and on click of view button below screen will be open and based this PRD TL/Hoss will review all planned marks (-->) and approve the request.

Line	Machine Name	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verified by (MTD TL)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
Line: M/c No : TEST-01	Machine Name: Test Machine	Ritu Bhatia Srekanth	Ashish Gupta	Approved by (MTD HOS) Approved by (MTD HOD)										
SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	Plt	Time change (min)					
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60						
2	B		Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75						
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120						
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90						
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60						
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40						
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60						

- If PRD TL/Hoss need to approve the request then select Yes and submit, else reject then select No and enter relatable remarks (required).

←

Kindly approve checkSheet. ☐ Yes ☒ No

Remarks:

Submit

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

Ashish Gupta

Year: 2023-2024

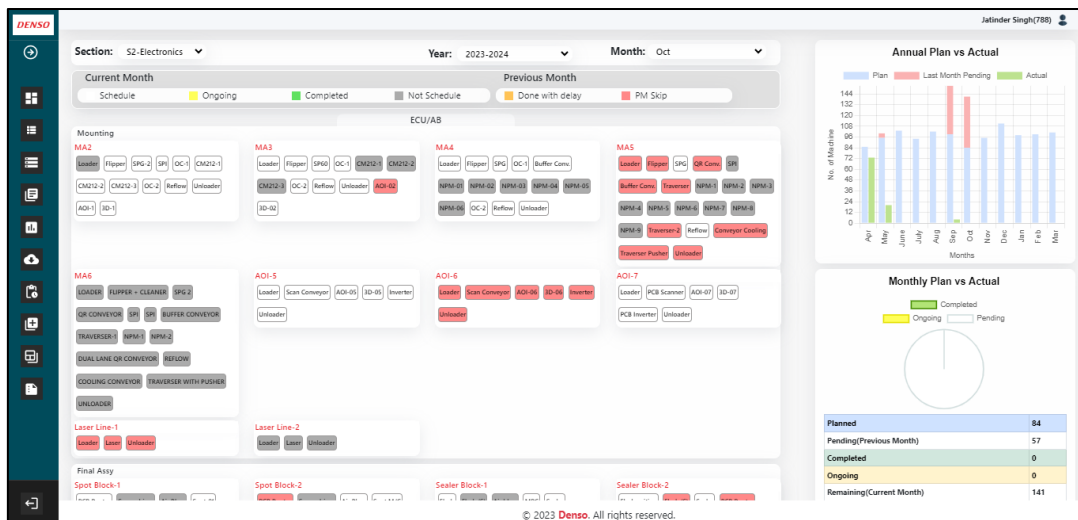
Line:- M/c No : TEST-01	Machine Name: Test Machine	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verify by (MTD TL)															
		Ritu Shadia .Sneekamth	Ashish Gupta	Approved by (MTD HOS)															
				Approved by (MTD HOD)															
SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person	In charge	PM Time charge (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M		60	-->			-->			-->			-->
2	B		Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M		75		-->			-->			-->		
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M		120			-->			-->			-->	
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M		90	-->			-->			-->			-->
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M		60	-->			-->			-->			-->
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M		40	-->			-->			-->			-->
				Smooth running	Rotation														

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3.2.12 Operator

3.2.12.1 Implementation phase work

- There is different color indication of the machine:
 - White -Indicate that machine is plan on current month and operator should work on this.
 - Gray - Indicate that machine is not plan/schedule on current month.
 - Yellow - Indicate that machine work is ongoing (half done / half remaining).
 - Green - Indicate that machine work is completed (need to send approval request to higher authority for work approval).
 - Orange - Indicate that machine work is done with delay (work not done in previous month but the remaining work done in current month).
 - Red - Indicate that machine planned work was skip.(not work in any month to the next frequency)



- Operator monitor every machine status and based on this user can start the work on selecting machine and below check-sheet will open.

CheckSheet

Close

PLAN ACCEPTANCE
(By PRD TL)

Harsh

PLAN PREPARED
(By MTD TL)

Ashish Gupta

Year: 2023-2024

Line:- Spot Block-1 M/c No : TEST-01	Machine Name: Test Machine	Approved by (MTD HOS) Ritu Bhatia .Srekanth	Prepared by (MTD TL) Ashish Gupta	Checked & Verify by (MTD TL)															
Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verify by (MTD TL)																	
SN	C	Inspection Item	Inspection point	Judgment criteria	Action	Cycle	Person in charge	Time (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60	-->			-->			-->			-->	
2	B		Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75		-->			-->			-->			-->
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120			-->			-->	-->		-->		
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90	-->			-->			-->			-->	
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60	-->			-->			-->			-->	
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40	-->			-->			-->			-->	
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60	-->	-->			-->			-->			-->
8	B		No dust on the LED	Clean by					-->			-->			-->			-->	

- When user will click on the planned button which is enable, user will show the above popup and start fill data as machine condition.

CheckSheet

Close

PLAN ACCEPTANCE
(By PRD TL)

Harsh

PLAN PREPARED
(By MTD TL)

Ashish Gupta

Year: 2023-2024

Line:- Spot Block-1 M/c No : TEST-01	Machine Name:	Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verify by (MTD TL)															
Approved by (MTD HOS)	Prepared by (MTD TL)	Checked & Verify by (MTD TL)																	
SN	C	Inspection Item	Inspection point	Judgment criteria	Action	Cycle	Person in charge	Time (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60	-->			-->			-->			-->	
2	B		Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75		-->			-->			-->			-->
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120			-->			-->	-->		-->		
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90	-->			-->			-->			-->	
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60	-->			-->			-->			-->	
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40	-->			-->			-->			-->	
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually	1/3M	M	60	-->	-->			-->			-->			-->
8	B		No dust on the LED	Clean by					-->			-->			-->			-->	

Work on Implementation

☒ OK
 ☐ Rectify
 ☐ NG (Not Good)

Remarks:

Photo Upload: No file chosen

Save

- On the above screen user will select OK and user will need to fill require fields and after click on save for store user work for future use.

CheckSheet

Close

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

March

Ashish Gupta

Year: 2023-2024

Oct	Nov	Dec	Jan	Feb
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->

Work on Implementation

☐ OK
 ☒ Rectify
 ☐ NG (Not Good)

Remarks:

Abnormality Details

Sr.No:

Abnormality Remarks:

Abnormality Status:

Photo Upload: No file chosen

Spare Details

Sr.No:

Part Name:

Part No:

Cost(INR):

Save

- If user will select Rectify, then add every details like abnormality details and spare details (if used).

CheckSheet

Close

PLAN ACCEPTANCE
(By PRD TL)

PLAN PREPARED
(By MTD TL)

Ashish Gupta

Year: 2023-2024

Oct	Nov	Dec	Jan	Feb
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->
-->	-->	-->	-->	-->

Work on Implementation

☐ OK
 ☒ Rectify
 ☐ NG (Not Good)

Remarks:

Abnormality Details

Sr.No:

Abnormality Remarks:

Abnormality Status:

Target Date:

Photo Upload: No file chosen

Spare Details

Sr.No:

Part Name:

Part No:

Cost(INR):

Save

- The user will select NG(Not good) when machine condition in abnormal and fill require details as shown in popup.

Close

Done By (MTD TM's)

Quality Check (By PRD TL)

Revised Plan Approved(MTD HOS)

Jatinder Singh

Sr. No.

Revision contents

Date

Revised by

Summary

[Notes of filing out checklist] (Category)

Breakdown: Directly relates to failure aspect

Safety: Directly relates to safety aspect

Quality: Directly relates to quality aspect

Pollution: Directly relates to pollution aspect

(Person in charge)

M: Maintenance personnel

O : Production personnel

Delay reason:

PM Status: Completed

PM Time(min):

Previous PM Time(min): 0

Supporting TM: Select

Save

PRD TL List (Quality Check)

MTD TL List (Checked & Verify by)

MTD HOS List (Approved by)

Please select

Please select

Please select

Send for Approval

--> Planned

→ Normal Condition

--> * Abnormality

FO/MTD/02/04/04

- **NOTE (For operator):** After completion of every points which planned in current month, user need add PM time, supporting TM(If worked with) compulsory to save.
- When completion of all points which is planned on current month, after the approval box is enable. Operator need select PRD TL, MTD TL and MTD HOS for sending approval.

Close

PLAN ACCEPTANCE (By PRD TL)

PLAN PREPARED (By MTD TL)

Harsh

Ashish Gupta

Year: 2023-2024

Line: Spot Block-1

M/c No : TEST-01

Machine Name: Test Machine

Approved by (MTD HOS)

Prepared by (MTD TL)

Checked & Verify by (MTD TL)

Approved by (MTD HOS)

Approved by (MTD HOD)

Ritu Bhatia .Srekanth

Ashish Gupta

SN	C	Inspection Item	Inspection point	Judgement criteria	Action	Cycle	Person in charge	PM Time (min)	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb
1	S	Area sensor	Area sensor operation check	Should give output	Check & replace	1/3M	M	60	-->			-->			-->			-->	
2	B		Dust collector box	No looseness in the dust collector box	Check & Fixed properly	1/3M	M	75		-->			-->			-->			-->
3	Q	Chamber unit	Dust collector	No damage & wear out the dust collector	Repair & replaced	1/3M	M	120			-->			-->	→ 2		-->		
4	Q		Packing	No leakage from chamber	Repair & replaced	1/3M	M	90	-->			-->			→ 2			-->	
5	B	Manometer	Pipe blockage check	No blockage inside the pipe line	Clean	1/3M	M	60	-->			-->			→			-->	
6	B		Water level check	Regular water level check	Supply or refill	1/3M	M	40	-->			-->			→			-->	
7	B	Spindle collet / Spindle motor	Movement checking	Smooth running without any abnormal sound	Rotation check manually.	1/3M	M	60		-->			-->			-->			-->

- If any of the higher authority reject approval request Operator need to update his/her work according to the rejected remarks which is given by higher authority.

CheckSheet

Line: Spot Block-1
M/c No : TEST-01

SN	C	Inspection
1	S	Area sensor
2	B	Chamber unit
3	Q	
4	Q	
5	B	Manometer
6	B	
7	B	Spindle collet / motor
8	B	Camera

Edit Remarks of month (Oct)

Sr. No.	Inspection Item	Remarks	Abnormality(Yes/No)	Ab. Remarks	Status	T.D	Spare(Yes/No)	P. Name	P. No	Cost	TM	Actions
1	Area sensor										Jatinder Singh	
2	Chamber unit										Jatinder Singh	
3	Chamber unit	2									Jatinder Singh	
4	Chamber unit	2									Jatinder Singh	
5	Manometer										Jatinder Singh	
6	Manometer										Jatinder Singh	
7	Spindle collet / Spindle motor										Jatinder Singh	
8	Camera										Jatinder Singh	
9	X,Y,Z -axis										Jatinder Singh	
10	X,Y,Z -axis										Jatinder Singh	
11	Air section										Jatinder Singh	

PRD TL List (Quality Check) Please select

MTD TL List (Checked & Verify by) Please select

MTD HOS List (Approved by) Please select

Send for Approval

Close

CheckSheet

Line: Spot Block-1
M/c No : TEST-01

SN	C	Inspection
1	S	Area sensor
2	B	Chamber unit
3	Q	
4	Q	
5	B	Manometer
6	B	
7	B	Spindle collet / motor
8	B	Camera

Rejected Remarks: Need to add remarks on point 1 and

Close

Edit Remarks of month (Oct)

Sr. No.	Inspection Item	Remarks	Abnormality(Yes/No)	Ab. Remarks	Status	T.D	Spare(Yes/No)	P. Name	P. No	Cost	TM	Actions
1	Area sensor	1									Jatinder Singh	✓ X
2	Chamber unit										Jatinder Singh	
3	Chamber unit	2									Jatinder Singh	
4	Chamber unit	2									Jatinder Singh	
5	Manometer										Jatinder Singh	
6	Manometer										Jatinder Singh	
7	Spindle collet / Spindle motor										Jatinder Singh	
8	Camera										Jatinder Singh	
9	X,Y,Z -axis										Jatinder Singh	
10	X,Y,Z -axis										Jatinder Singh	

Close

- **On rejection:** For updating rejected work Operator will see the red button enable on the rejected month header, click on this button one popup is open and Operator will able edit points based on given remarks and on the same popup approval box is enable for again need to send approval to the every higher authority user.
- Operator user can see the approver name when approver will approve the receive the request.

3.2.13 Mid-year Check-sheet edition

- The machine check-sheet will be edit on mid-year on check-sheet dashboard and this edition done by MTD TL/Hoss or Operator.

Sr. No.	Cell/Product	Line	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-1	EEAT-094	PCB router-01	View Edit Delete
2	Final Assy	Spot Block-1	EEAT-034	Screw driver-04	View Edit Delete
3	Final Assy	Spot Block-1	EEAT-096	Spot-01	View Edit Delete
4	Final Assy	Spot Block-1	EEAT-026	AOI-04	View Edit Delete
5	Final Assy	Spot Block-1	EEAT-028	3D-04	View Edit Delete
6	Final Assy	Spot Block-1	EEAT-149	Air Blowing M/C	View Edit Delete
7	Final Assy	Spot Block-1	TEST-01	Test Machine	View Edit Delete

- When user will click on the edit button the below screen will be open and user will able to add/ update/ delete check-sheet points as require.

3.2.13.1 On MTD TL/Hoss and Operator

Sr. No.	C	Inspection Item	Inspection Point	Judgement Criteria	Action	Cycle	Person In Charge	PM Time (min)	Actions
1	B	PCB Flipper	Conveyor belt	Tension/wear out	Clean/Replace/Adjust	1/1M	M	30	Edit Delete
2	B	Vacuum Cleaner	Working Check	Should work properly without abnormal Sound	Clean/Replace/Adjust	1/1M	M	15	Edit Delete
3	B	Pneumatic section	Filter element	There should be no dirt/dust on the filter element & bowl. Check/Clean/Replace	Check/Clean/Replace	1/3M	M	10	Edit Delete
4	B	Pneumatic section	All Cylinder	Should be smooth, No jerking & abnormal sound	Check/Replace	1/3M	M	60	Edit Delete
5	B	Cooling Fan	Movement	Should run without any abnormal noise	Check/Replace	1/3M	M	45	Edit Delete
6	B	Sensor	Condition	There should be no oil or dirt on optical side surface	Clean/Replace	1/3M	M	45	Edit Delete
7	B	Each section cable	Condition	No peel Off or cut marks on insulation	Check/Replace	1/6M	M	20	Edit Delete
8	S	Safety device	Safety switch	M/c Should be stop immediately when activated	Repair or Replace	1/6M	M	30	Edit Delete

- Note:** The user will take any action(add/ update/ delete) on check-sheet, compulsory need to add every action taken detail on the revision content table. If not add it will not permission for further process of approval and check-sheet implementation.

The screenshot shows the Denso Predictive Maintenance System interface. A table lists maintenance tasks with columns for Sr. No., Revision contents, Date, Revised by, and Actions. A warning message is displayed: "Please fill all revision content which you (ADD/UPDATE/DELETE)!" with a red border. The table contains 19 rows of data. Below the table, there is a search bar and a table with columns: Sr. No., Revision contents, Date, Revised by, and Actions. The table is currently empty, showing "No records to display".

Sr. No.	Revision contents	Date	Revised by	Actions
No records to display				

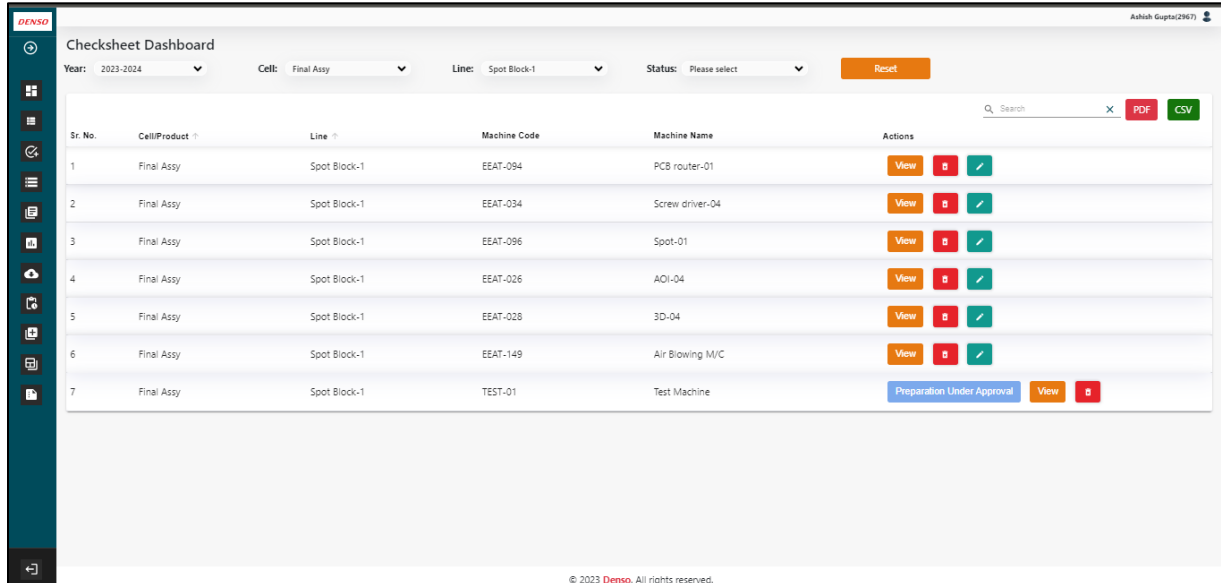
- When user not entered any revision while mid-year change it throws the warning.

The screenshot shows the Denso Predictive Maintenance System interface. A table lists maintenance tasks with columns for Sr. No., Revision contents, Date, Revised by, and Actions. A warning message is displayed: "Please fill all revision content which you (ADD/UPDATE/DELETE)!" with a red border. The table contains 19 rows of data. Below the table, there is a search bar and a table with columns: Sr. No., Revision contents, Date, Revised by, and Actions. The table contains one row of data: 1, add new point Testing, 2023-10-10, Ashish Gupta, and a trash icon.

Sr. No.	Revision contents	Date	Revised by	Actions
1	add new point Testing	2023-10-10	Ashish Gupta	

- After adding revision content it will give permission for further process.
- After user will able to click on view and send for approval for sending approval to the higher authority and check sheet status will go on the "Preparation" stage.
- MTD TL/Hoss or Operator need to remember that when user edit cycle of the particular point need change start month on "Planning" phase. If user select as previously planned start month then the previously work is remain same data.

- When user delete any points of check-sheet from already created, the points will be highlighted as red and when new financial year will come it will not display on the machine check-sheet.



Checksheet Dashboard

Year: 2023-2024 Cell: Final Assy Line: Spot Block-1 Status: Please select Reset

Search PDF CSV

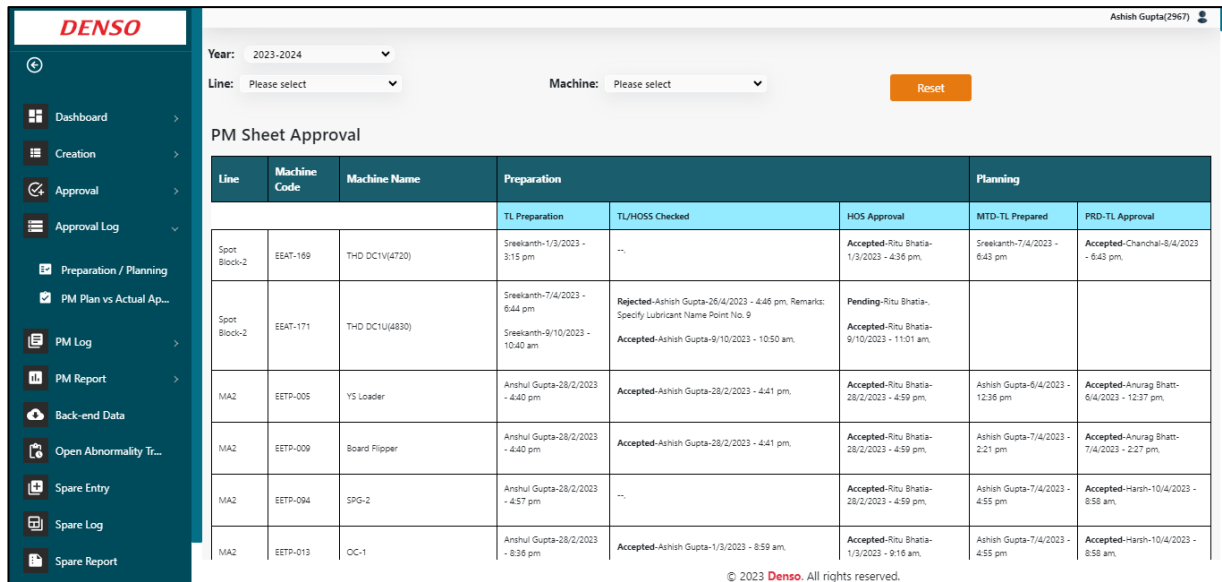
Sr. No.	Cell/Product	Line	Machine Code	Machine Name	Actions
1	Final Assy	Spot Block-1	EEAT-094	PCB router-01	View PDF CSV
2	Final Assy	Spot Block-1	EEAT-034	Screw driver-04	View PDF CSV
3	Final Assy	Spot Block-1	EEAT-096	Spot-01	View PDF CSV
4	Final Assy	Spot Block-1	EEAT-026	AOI-04	View PDF CSV
5	Final Assy	Spot Block-1	EEAT-028	3D-04	View PDF CSV
6	Final Assy	Spot Block-1	EEAT-149	Air Blowing M/C	View PDF CSV
7	Final Assy	Spot Block-1	TEST-01	Test Machine	Preparation Under Approval View PDF

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- On mid-year edition the work flow will same go on like “Preparation -> send approval of preparation -> approve by higher authorities -> Planning -> send approval of planning -> approve by higher authority -> Implementation -> work on check-sheet by Operator”.

3.2.15 Log/History Dashboards

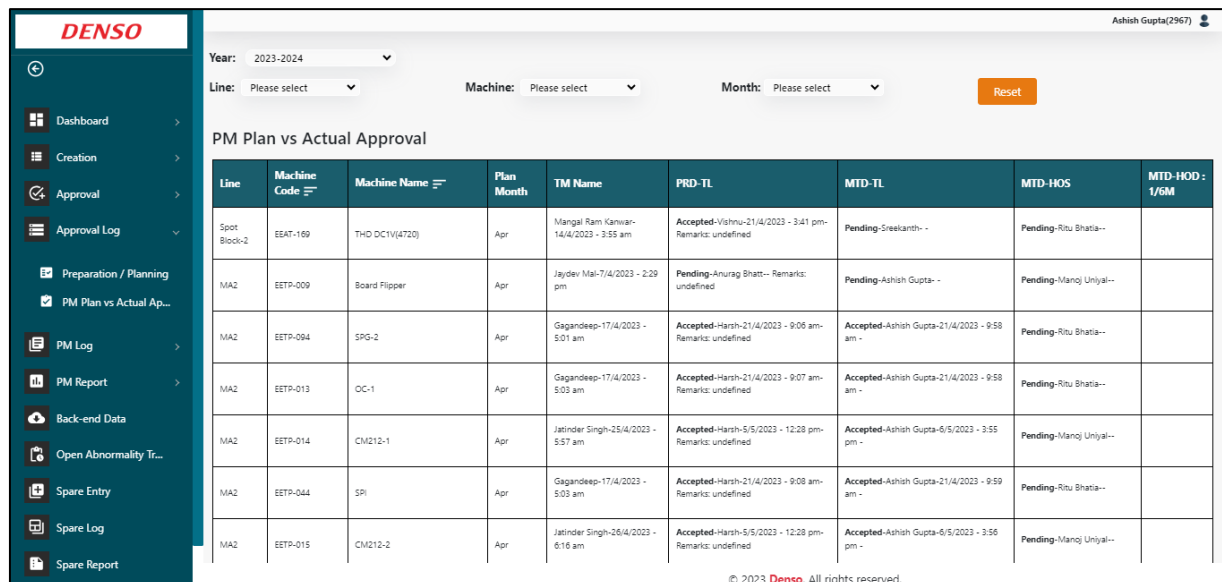
3.2.15.1 Preparation/Planning approval log



Line	Machine Code	Machine Name	Preparation	Planning
			TL Preparation	TL/HOSS Checked
			HOS Approval	MTD-TL Prepared
			PRD-TL Approval	
Spot Block-2	EEAT-169	THD DC1V(4720)	Sreelanth-1/3/2023 - 3:15 pm	Accepted-Ritu Bhatia-1/3/2023 - 4:36 pm
Spot Block-2	EEAT-171	THD DC1V(4830)	Sreelanth-7/4/2023 - 6:44 pm Sreelanth-8/10/2023 - 10:40 am	Rejected-Ashish Gupta-26/4/2023 - 4:46 pm. Remarks: Specify Lubricant Name Point No. 9 Accepted-Ritu Bhatia-9/10/2023 - 11:01 am
MA2	EETP-005	YS Loader	Anshul Gupta-28/2/2023 - 4:40 pm	Accepted-Ritu Bhatia-28/2/2023 - 4:59 pm
MA2	EETP-009	Board Flipper	Anshul Gupta-28/2/2023 - 4:40 pm	Accepted-Ashish Gupta-28/2/2023 - 4:41 pm
MA2	EETP-094	SPG-2	Anshul Gupta-28/2/2023 - 4:57 pm	Accepted-Ritu Bhatia-28/2/2023 - 4:59 pm
MA2	EETP-013	OC-1	Anshul Gupta-28/2/2023 - 8:36 pm	Accepted-Ashish Gupta-1/3/2023 - 8:59 am

- On this dashboard every user will see the history of every preparation and planning phase approval log. Also, user can filter out data based on year, line and machine.

3.2.15.2 PM Plan vs Actual (Implementation Phase) Approval log



Line	Machine Code	Machine Name	Plan Month	TM Name	PRD-TL	MTD-TL	MTD-HOS	MTD-HOD : 1/GM
Spot Block-2	EEAT-169	THD DC1V(4720)	Apr	Mangal Ram Kanwar-14/4/2023 - 3:55 am	Accepted-Vishnu-21/4/2023 - 3:41 pm-Remarks: undefined	Pending-Sreelanth--	Pending-Ritu Bhatia--	
MA2	EETP-009	Board Flipper	Apr	Jaydev Mali-7/4/2023 - 2:29 pm	Pending-Anurag Bhatt--Remarks: undefined	Pending-Ashish Gupta--	Pending-Manoj Unjyal--	
MA2	EETP-094	SPG-2	Apr	Gagandeep-17/4/2023 - 5:01 am	Accepted-Harsh-21/4/2023 - 9:06 am-Remarks: undefined	Accepted-Ashish Gupta-21/4/2023 - 9:58 am -	Pending-Ritu Bhatia--	
MA2	EETP-013	OC-1	Apr	Gagandeep-17/4/2023 - 5:03 am	Accepted-Harsh-21/4/2023 - 9:07 am-Remarks: undefined	Accepted-Ashish Gupta-21/4/2023 - 9:58 am -	Pending-Ritu Bhatia--	
MA2	EETP-014	CM212-1	Apr	Jatinder Singh-25/4/2023 - 5:57 am	Accepted-Harsh-5/5/2023 - 12:28 pm-Remarks: undefined	Accepted-Ashish Gupta-5/5/2023 - 3:55 pm -	Pending-Manoj Unjyal--	
MA2	EETP-044	SPI	Apr	Gagandeep-17/4/2023 - 5:03 am	Accepted-Harsh-21/4/2023 - 9:08 am-Remarks: undefined	Accepted-Ashish Gupta-21/4/2023 - 9:59 am -	Pending-Ritu Bhatia--	
MA2	EETP-015	CM212-2	Apr	Jatinder Singh-26/4/2023 - 6:16 am	Accepted-Harsh-5/5/2023 - 12:28 pm-Remarks: undefined	Accepted-Ashish Gupta-5/5/2023 - 3:56 pm -	Pending-Manoj Unjyal--	

- User will see Plan vs Actual approval log on this dashboard. Also they filter out data based on year, line, machine, and month.

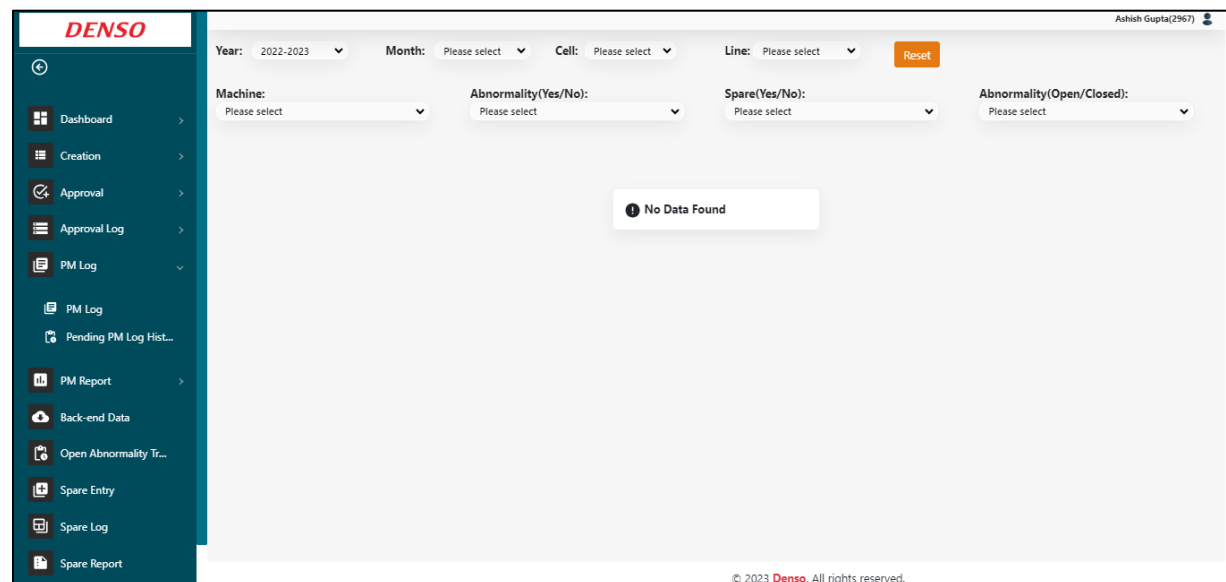
3.2.15.3 PM Logs

DENSO

</

- On PM log dashboard user will see every data of machine check-sheet (with every inspection point).
- Also give additional filter so that user can filter out data easily from the lots of data. User can download images which will be uploaded by Operator.

3.2.15.4 Pending PM Log History



Pending PM Log History																			
No Data Found																			

- On this dashboard user will see data of skip and pending machine log which will be done on any incoming month.
- Also give additional filter so that user can filter out data easily from the lots of data. User can download images which will be uploaded by Operator.

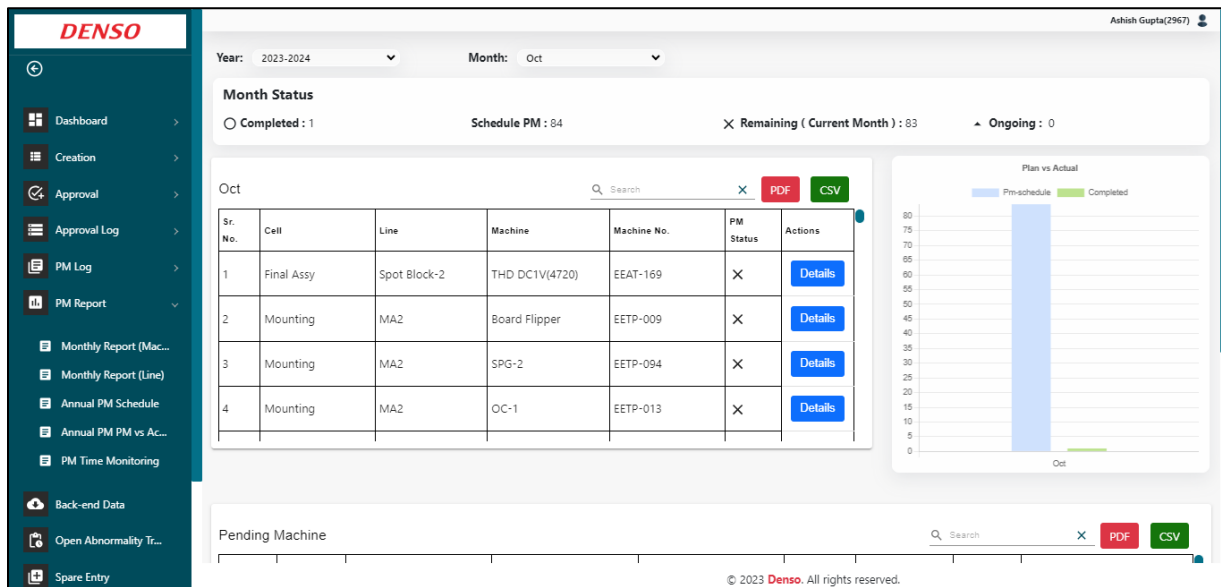
3.2.15.5 Spare Log

Sr. No.	Date	Line	Machine	Machine No.	Category	Part Name	Part No.	Used By	Cost	Abnormality	SparePart	Action
									28915.96			
1	11/4/2023 - 4:55 am	MA4	OC-2	EETP-058	PM	BELT	HNB-5E 11Wx114EEF	Dalip Khajuria	4900	Yes	Yes	
2	2/5/2023 - 5:18 am	AOI-6	3D-06	EETP-090	PM	Belt	HNB-5E 14 2.5Wx1600EF	Dalip Khajuria	1640	Yes	Yes	
3	2023-05-01	FA-WP3	Silicon Below connector	EEAT-153	Corrective	Spring	PU-01020-00-4	Sandeep Dhama	363	No		
4	11/4/2023 - 3:32 am	Spot Block-2	Screw driver-02	EEAT-032	PM	CARBON BRUSH FOR SCREW DRIVER	TD00579-0	Dalip Khajuria	200	Yes	Yes	
5	2023-04-01	Spot Block-1	PCB router-01	EEAT-094	BM	sdg	57752	Ashish Gupta	5732	No		
6	2023-05-07	Spot Block-1	Spot-01	EEAT-096	BM	MAGNETIC CONTACTOR	SC-01 AC200V 18	Subash Sheoran	1475.96	No		
7	2023-05-07	Spot Block-1	Spot-01	EEAT-096	BM	PROXIMITY SENSOR	E2EM-V4X2 2M	Subash Sheoran	3002	No		
8	2023-05-07	Spot Block-1	Spot-01	EEAT-096	BM	ABS BATTERY	WSB-5	Subash Sheoran	1875	No		

- User will display logs of every spare entry on this dashboard with PM, BM, CM, Kaizen.
- Also user can apply filter as his/her requirement like year change, category wise, month, line machine.
- User will also able to show total cost of overall spare usage.
- MTD TL/Hoss and Operator will able to delete this entry if needed.

3.2.16 PM Reports

3.2.16.1 Monthly Reports (Machine wise)



- User can show current month data with completed, schedule, remaining and ongoing machine count.
- Also, User will show and compare Plan vs actual machine on charts, based on this user will take decision easily.
- Below screen represent that user will show previous month no completion and skip machines list. User will change the month and based on this display selected month data and it's respected previous month pending data.
- For updating skip machine need to select completion target date and after send approval and then PM Edit button will enable.
- User will able to complete the skip machine when click on the PM Edit button which is open new popup, which is same as when worked on implementation data. If user complete skip machine done with delay status it will display up-to one month on this dashboard after it will not show.
- And this log will be display on pending PM log dashboard.
- On every month of first, mid and last day one email will be generated and send to the higher authority for review and take decision.

Pending Machine

Sr. No.	Schedule Month	Cell	Line	Machine	Machine No.	Completion Target Date	PM Status	Actions
1	Sep	Mounting	MA2	YS Loader	EETP-005		No Completion	Details
2	Sep	Mounting	MA3	AOI-02	EETP-004		No Completion	Details
3	Sep	Mounting	MA5	YS Loader	EETP-073		No Completion	Details
4	Sep	Mounting	MA5	Board Flipper and Cleaner	EETP-074		No Completion	Details

MTD HOS : Ritu Bhatia Status : **Pending**
MTD HOD : Yogesh Dhyani Status : **Pending**
PRD HOS : Rajeev Gupta Status : **Pending**
PRD HOD : Anup Shekhar Status : **Pending**
Reason for delay : Last Month Status OK. Software o...
Rejected remarks :

3.2.16.2 Monthly Report (Line wise)

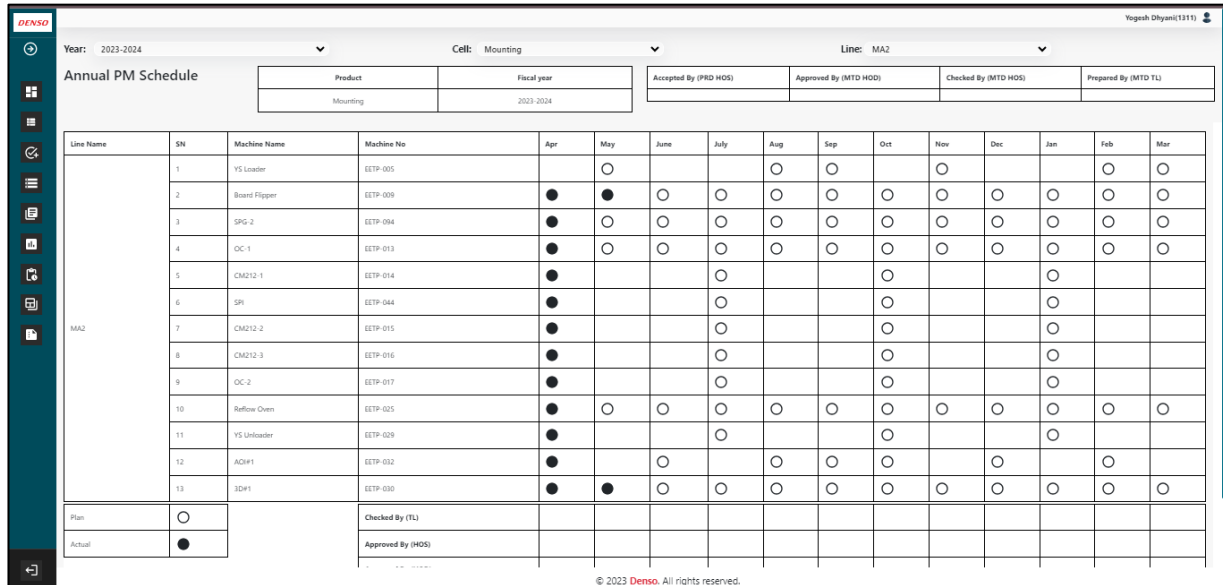
Year: 2023-2024 **Month:** Oct **Line:** Please select

Reset

Sr No	Line	PM Schedule	Last Month Pending	Completed	Pending
1	MA2	12	1	0	12
2	Spot Block-1	4	1	1	3
3	Meter 1	3	2	0	3
4	RTS-1	1	0	0	1
5	MA3	8	1	0	8
6	Spot Block-2	4	3	0	4
7	Meter 3	3	1	0	3
8	RTS-2	1	2	0	1
9	MA4	8	0	0	8
10	Sealer Block-1	3	0	0	3
11	MA5	2	9	0	2
12	Sealer Block-2	2	2	0	2
13	Sealer Block-3	3	0	0	3
14	AOI-5	6	0	0	6
15	F&WDP	1	1	0	1

- User will be able to see data line wise and also with PM schedule, last month pending, completed, pending counts.
- On click of every count user can see machine list according to status count selection.
- Also user will see data on charts and take decision quickly.

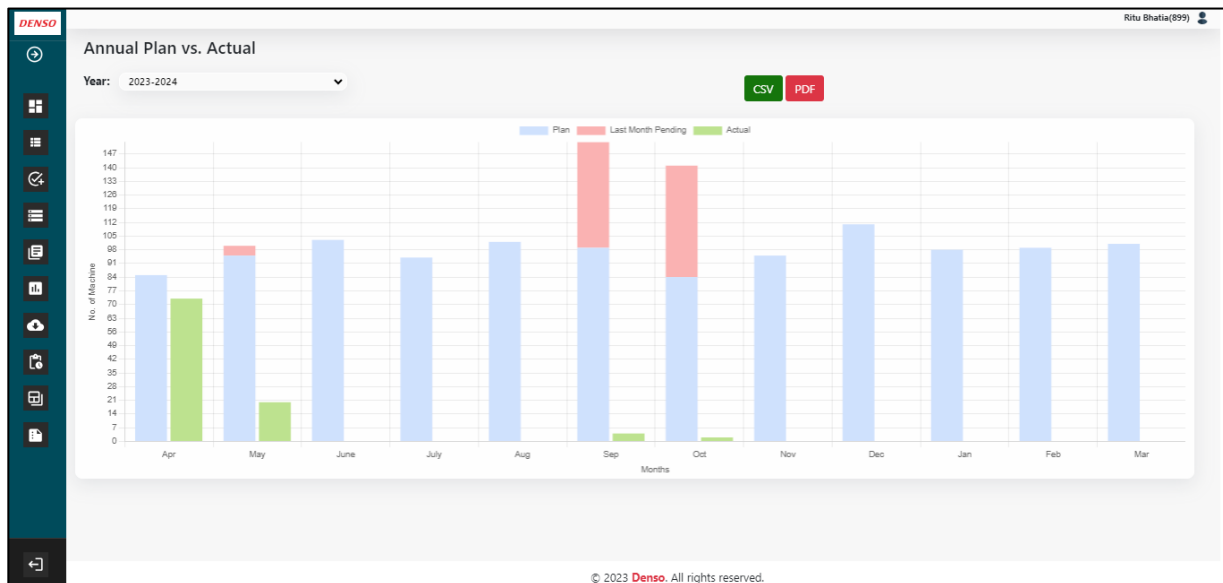
3.2.16.3 Annual PM Schedule



Line Name	SN	Machine Name	Machine No	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
MA2	1	YS Loader	ETEP-005		○			○	○		○			○	○
	2	Board Flipper	ETEP-009	●	●	○	○	○	○	○	○	○	○	○	○
	3	SPG-2	ETEP-004	●	○	○	○	○	○	○	○	○	○	○	○
	4	OC-1	ETEP-013	●	○	○	○	○	○	○	○	○	○	○	○
	5	CM212-1	ETEP-014	●			○			○			○		
	6	SPI	ETEP-044	●			○			○			○		
	7	CM212-2	ETEP-015	●			○			○			○		
	8	CM212-3	ETEP-016	●			○			○			○		
	9	DC-2	ETEP-017	●			○			○			○		
	10	Reflow Oven	ETEP-025	●	○	○	○	○	○	○	○	○	○	○	○
	11	YS Unloader	ETEP-029	●			○			○			○		
	12	JQH1	ETEP-032	●		○		○	○	○		○		○	
	13	JDH1	ETEP-030	●	●	○	○	○	○	○	○	○	○	○	○

- User can show month wise plan vs actual with dark and light circle symbol. So, user can take decision easily.
- Also, User will able to apply filter for other cell and line wise annual pm schedule for the selected cell and line. User will see the approval of higher authority.

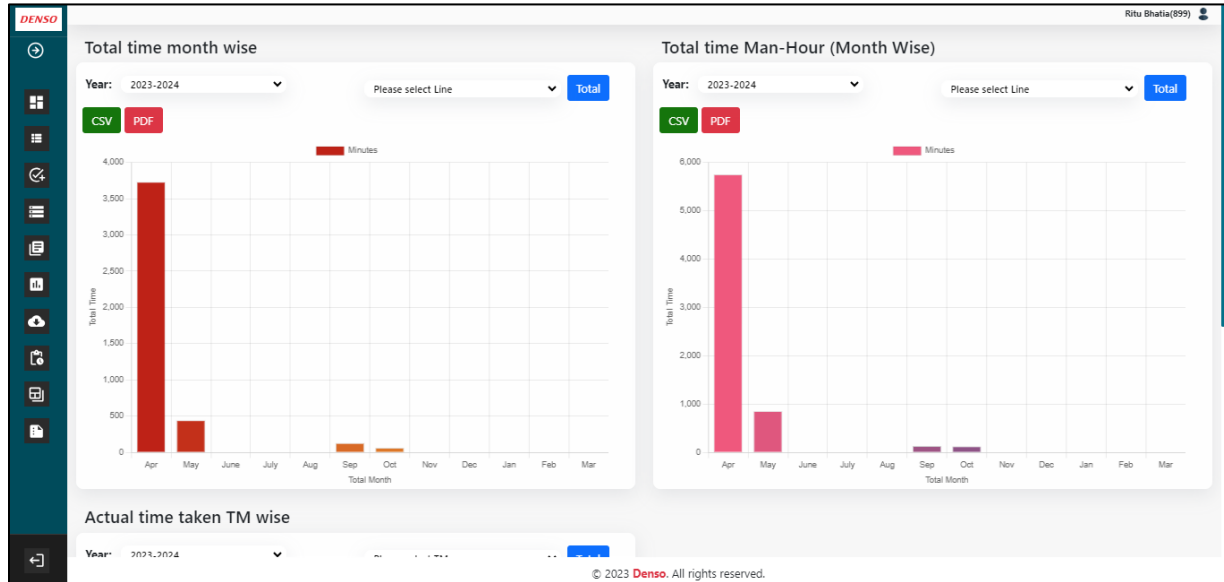
3.2.16.4 Annual Plan vs Actual



- User can see the plan vs actual of the particular assign section as year wise.

- User will be able to show past year data and download data as PDF and CSV.

3.2.16.5 PM Time Monitoring



- The PM time monitoring dashboard is basically display TM(Team member) work hour(in minutes).
- Also this dashboard contain different type of data charts like:
 - Total time month wise: This indicates overall time and after user will be able to show line wise data based on line selection.
 - Total time man-hour(month wise) : This include every hour of supporting team member selection while work on implementation.
 - Actual time taken TM wise: This contain overall time and also user can see total time taken by the particular TM(Team member) based on selection.

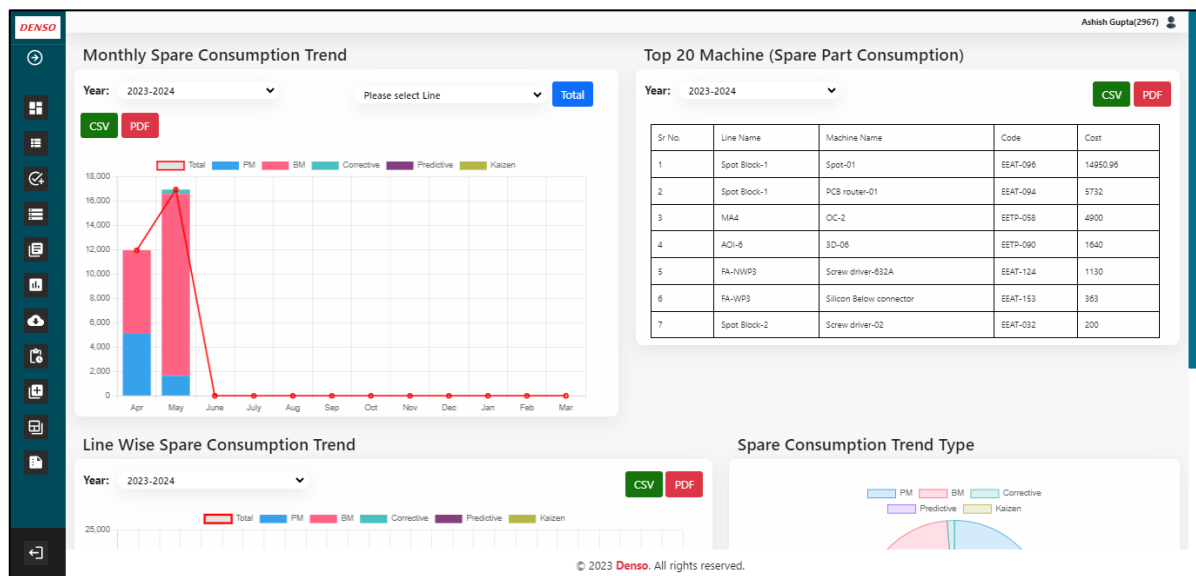
3.2.17 Back-end Data Dashboard

Sr No	Line Name	Machine Code	Machine Name	Actions
1	FP-Armature	RM-EN-FP-ARM-1-010-1	SHAFT PRESS FIT	View Delete
2	MA2	REETP-005	YS Loader	View Delete
3	MA2	REETP-009	Board Flipper	View Delete
4	Spot Block-2	REEAT-171	THD DC1U(4830)	View Delete
5	MA2	REETP-094	SPG-2	View Delete
6	Cell Sensor	REEAT-182	PCB Router	View Delete
7	MA2	REETP-013	OC-1	View Delete
8	MA2	REETP-014	CM212-1	View Delete

- User can show deleted machine check-sheet data also they will use in new check-sheet which is same as deleted data on check-sheet dashboard.
- Only MTD Hos user will delete this back-end data.

3.2.18 Spare Reports

- This dashboard contain four charts and each chart it's own different data analysis.



- Monthly spare consumption trend :

3.2.19 Profile

- User will be able to update their name and profile image on this dashboard.

- Also, user will be able to update their password. Need to add first old password then new.

3.2.20 Common Approval of Reports

3.2.20.1 Annual PM Schedule (MTD TL, MTD HOS, MTD HOD, PRD HOS)

Annual PM Schedule

Year: 2023-2024 Cell: Mounting Line: MA2

Product	Fiscal year	Accepted By (PRD HOS)	Approved By (MTD HOD)	Checked By (MTD HOS)	Prepared By (MTD TL)
Mounting	2023-2024	Please select	Please select	Please select	Send For Approval

Line Name	SN	Machine Name	Machine No	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
MA2	1	YS Loader	EETP-005												
	2	Board Ripper	EETP-009												
	3	SPG-2	EETP-064												
	4	OC-1	EETP-013												
	5	CM212-1	EETP-014												
	6	SPI	EETP-044												
	7	CM212-2	EETP-015												
	8	CM212-3	EETP-016												
	9	OC-2	EETP-017												
	10	Reflow Oven	EETP-025												
	11	YS Unloader	EETP-029												

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Note:

- On this dashboard first approval will send by MTD TL/Hoss and selected higher authority for approve.
- After MTD HOS will receive mail regarding the cell and line based annual PM schedule list. The MTD HOS need to go approval dashboard and select cell and line which is mentioned in email for approval and then click on the approve button after MTD HOD will receive the mail regarding approval.
- MTD HOD will approve annual PM schedule based on selected cell/products and line machines on mention in email.
- After approval of MTD HOD approval, PRD HOS will receive the mail regarding approval and approve the request.
- By default first cell and line display on dashboard.
- You need to change cell and line value base on receive approval email.
- Every selected user will receive mail one after one approval MTD HOS, MTD HOD, PRD HOS will receive email for approval of annual PM schedule.
- The approve button will enable on the approved by (MTD HOD) column, after the MTD HOS is check.

- After approval of (MTD HOS, MTD HOD, PRD HOS) line wise, the below approval is enable in each month.

- Home
- Dashboard
- Calendar
- Reports
- Settings
- Profile
- Logout

MA2

4	OC-1	EETP-013	●	○	○	○	○	○	○	○	○	○	○	○	○
5	CM212-1	EETP-014	●			○			○			○			
6	SPI	EETP-044	●			○			○			○			
7	CM212-2	EETP-015	●			○			○			○			
8	CM212-3	EETP-016	●			○			○			○			
9	OC-2	EETP-017	●			○			○			○			
10	Reflow Oven	EETP-025	●	○	○	○	○	○	○	○	○	○	○	○	○
11	YS Unloader	EETP-029	●			○			○			○			
12	AOI#1	EETP-032	●		○		○	○	○		○		○		
13	3D#1	EETP-030	●	●	○	○	○	○	○	○	○	○	○	○	○
Plan	○	Checked By (TL)													
Actual	●	Approved By (HOS)													
		Approved By (HOD) (Only in case of delay)													
		Remarks (If Delay)													

FO/MTD/02/18/00

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- After click on edit button is give the option delay yes/no and select MTD HOS, If select Yes then select MTD HOD from the given list and then add remarks and send for approval.
- If any delay on completion of plan vs actual HOD will receive approval and add remarks on this.
- As users apply the filter, users can also examine past year and previous month data.
- Every approval is display on annual PM schedule dashboard.

3.2.20.2 Monthly Report (Machine wise) Approval (MTD HOS, MTD HOD, PRD HOS, PRD HOD)

DENSO

Pending Machine

Sr. No.	Schedule Month	Cell	Line	Machine	Machine No.	Completion Target Date	PM Status	Actions
1	Sep	Mounting	MA2	YS Loader	EETP-005		No Completion	Details
2	Sep	Mounting	MA3	AOI-02	EETP-004		No Completion	Details
3	Sep	Mounting	MA5	YS Loader	EETP-073		No Completion	Details
4	Sep	Mounting	MA5	Board Flipper and Cleaner	EETP-074		No Completion	Details

MTD HOS : Ritu Bhatia
Status : Pending

MTD HOD : Yogesh Dhyani
Status : Pending

PRD HOS : Rajeev Gupta
Status : Pending

PRD HOD : Anup Shekhar
Status : Pending

Reason for delay : Last Month Status OK. Software o...

Rejected remarks :

Kindly approve skipped PM Machines. ☐ Yes ☐ No

[Submit](#)

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Figure 6: Monthly Report (Machine wise) Approval

- Monthly report (Machine wise) approval dashboard will enable approval of pending machine list.
- First user will review pending machine list and then approve the whole list.
- This approval will enable in below of the pending machine list.
- On this pending list, first approval will send by MTD TL/Hoss and selected higher authority for approve.
- First send approval to the MTD HOS and receive mail regarding the approval. After, approval of MTD HOS, MTD HOD will receive mail and approval request on monthly report (machine wise) dashboard.
- If any higher authority reject this approval then need to add rejected remarks and it will display on dashboard, and then again send approval after review.
- The flow of approval go as mentioned above the remaining PRD HOS and PRD HOD will receive approval.
- Pending machine list contain skip and no completion machines list of the previous months.

3.2.21 Need to Know

- Not every user able to access every dashboard mention in user manual.

- On section dashboard user will only able to show assign section data not every plant section's.
- Every approval on PM should be done one after one and also receive the mail regarding the every approval.
- Operator need to click on save button while work on Implementation work and after completion of planned work of the month, need to enter PM time and supporting TM and then click on save. If Operator not save the work, the operator name not display on check-sheet and this will go on Operator will not work on planned check-sheet.
- Reports approval will display on respected report dashboard.
- Every new financial year MTD HODs of every plant need to click on enabled Update (Financial year) button. After one time click it will disappear and display again when new year will come.
- On implementation phase if any approval reject from higher authority then Operator need to visit check-sheet and the one red button enable on the month row and then edit and approve from opened pop-up.
- Every year on September and March month auto email will be generated for the six month approval for the MTD HOD side. This approval request display on six-month approval dashboard.

4 APPLICATION DEPLOYMENT INSTRUCTIONS

Kindly follow the below steps to run the application:

- Double-click on batchFileDENSO_PM.bat file, the project will run.
- Note : If user need to stop the running project, press CTRL+C on opened command prompt.
- If user need to change server configuration then first take backup to the any particular folder and open command prompt and type below commands:

Backup or restore step:

1) Take dump of the database (Backup / Export):

a) `mongodump --host=localhost --port=27017 --db=DENSO-PM`

2) Restore database from the dump:

a) Redirect up-to dump folder and then open command prompt and write this command:

i. `mongorestore dump`

Setup (Installations) required:

- Node version: 16.13.2
- NPM version: 8.1.2
- MongoDB version: 4.2.20